

The critical dimension for one-phase Bernoulli minimizers

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Abstract

We prove that seven is the critical dimension for the one-phase Bernoulli problem: every nonzero one-homogeneous global minimizer in dimensions at most six is flat, while a nonflat one-homogeneous global minimizer exists in dimension seven. It follows that the interior free boundary of a local minimizer is smooth in dimensions at most six. In dimension $n \geq 7$, its singular set has Hausdorff dimension at most $n - 7$, and this bound is sharp. In dimension seven, the singular set is locally finite.

1 Introduction

For a nonnegative function $v \in H^1(B)$, where $B \subset \mathbb{R}^d$ is a ball, the one-phase Bernoulli energy is

$$J(v; B) = \int_B (|\nabla v|^2 + \mathbf{1}_{\{v>0\}}) dx. \quad (1.1)$$

A nonnegative $u \in H_{\text{loc}}^1(\mathbb{R}^d)$ is a *global minimizer* if $J(u; B) \leq J(v; B)$ for every ball B and every nonnegative $v \in H^1(B)$ with $v - u \in H_0^1(B)$. The competitors in these comparisons are arbitrary Sobolev functions with the prescribed trace. A function is *one-homogeneous* if $u(rx) = ru(x)$ for every $r > 0$, almost everywhere in x . The nonzero flat solutions are

$$u(x) = (x \cdot e)_+, \quad |e| = 1.$$

Let d_* be the least dimension containing a nonzero, nonflat, one-homogeneous global minimizer.

Theorem 1.1. *Every nonzero one-homogeneous global minimizer of (1.1) in $\mathbb{R}^d$, $1 \leq d \leq 6$, is flat. There is a nonflat one-homogeneous global minimizer in $\mathbb{R}^7$. Consequently,*

$$d_* = 7.$$

A nonnegative $u \in H_{\text{loc}}^1(D)$ is a local minimizer in an open set $D \subset \mathbb{R}^n$ if it satisfies the same comparison on every ball compactly contained in D . At an interior free-boundary point x_0 , a *blowup* is a locally uniform limit of $u(x_0 + rx)/r$ along radii tending to zero. A point is regular if one such limit is a flat solution, and singular otherwise. Write $\text{Sing}(u)$ for the set of singular points in D .

Corollary 1.2 (Sharp singular-set bound). *Let $n \geq 2$, let $D \subset \mathbb{R}^n$ be open, and let $u \in H_{\text{loc}}^1(D)$ be a nonnegative local minimizer of the one-phase Bernoulli energy in D . Its interior free boundary is smooth outside $\text{Sing}(u)$. This set is empty when $n \leq 6$, is locally finite when $n = 7$, and satisfies*

$$\dim_{\text{H}} \text{Sing}(u) \leq n - 7 \quad (n \geq 7).$$

For every $n \geq 7$, a global minimizer attains this dimension bound.

The upper bound is the standard dimension-reduction consequence of the critical dimension, and sharpness follows by taking Cartesian products of the De Silva–Jerison cone with Euclidean space. We verify these deductions in Section 8.

The variational theory of the one-phase Bernoulli problem begins with Alt and Caffarelli’s existence and regularity theory [1]. Weiss’s monotonicity formula and dimension reduction [18] connect the singularities of general minimizers to homogeneous global minimizers; see also [17, Theorem 1.4 and Chapters 9–10]. Thus the first dimension admitting a nonflat minimizing cone determines both the threshold for singular free boundaries and the codimension bound furnished by this theory. Our main result answers positively the critical-dimension question whether $d_* = 7$.

A second strand of the theory passes from weak free boundaries to classical ones. Caffarelli’s Harnack-inequality approach established regularity for Lipschitz free boundaries [2]. De Silva developed a different improvement-of-flatness argument allowing a nonzero right-hand side [4, Theorems 1.1–1.2]; this is the approach presented in the regularity chapters of [17]. Higher regularity has classical antecedents in Kinderlehrer and Nirenberg [14, Theorem 2]. The version used here, for Lipschitz viscosity solutions, is due to De Silva, Ferrari and Salsa [6, Theorem 1.1 and Corollary 1.2]. These results make a classification of homogeneous minimizing blowups applicable to general local minimizers.

As recalled by Jerison and Savin [13, Introduction], the dimension-three rigidity result for stable homogeneous solutions with smooth cross-section is due to Caffarelli, Jerison and Kenig [3]. Jerison and Savin proved that stable homogeneous solutions with smooth cross-section are flat in dimensions at most four [13, Theorem 1.1]; their regularity consequence for minimizers [13, Corollary 1.2] gives $d_* \geq 5$. Their argument constructs functions of the Hessian satisfying inequalities for the linearized free-boundary problem. In the other direction, De Silva and Jerison constructed a nonflat homogeneous global minimizer in dimension seven [7, Theorem 1.1], proving $d_* < 7$. The minimizing property in that construction is with respect to all Sobolev competitors with the prescribed boundary trace. Firester, Tsiamis and Wang also construct further symmetric minimizing one-phase cones, including a second example in dimension seven [11, Theorem 1.4].

Recent work further describes cones with symmetry. Firester, Tsiamis and Wang [10, Theorem 1.1] prove that their cohomogeneity-one cones with $O(n - k) \times O(k)$ symmetry, $1 \leq k \leq n - 2$, are unstable for $n \leq 6$ and strictly stable for $n \geq 7$. The exclusion proved here applies to globally minimizing cones without a symmetry assumption.

Cone classification also differs from rigidity of arbitrary entire stable solutions. Fernández-Real and Serra prove that every global classical stable solution in dimension four is one-dimensional [8, Theorem 1.1]. Their result removes homogeneity in that dimension.

For an admissible increasing family of nonnegative boundary data, Fernández-Real and Yu prove regularity through dimension six for almost every parameter [9, Assumption 1 and Theorem 1.3]. This is a generic-data result: the increase is required at a fixed linear rate on the positive set of the smaller boundary datum.

The obstacle and the proof strategy. The stability inequality compares the Dirichlet energy of a test function with a curvature term on the free boundary. The use of the harmonic Hessian has a geometric precedent in Simons’s second-fundamental-form argument for minimal cones [16, Section 6.1], a connection developed explicitly by Jerison and Savin. A natural Bernoulli test is a power of the Hessian norm. Jerison and Savin explain why this unweighted choice does not give a uniform instability criterion in dimensions four through six: the interior and boundary estimates impose different restrictions on the power, and their sharp configurations need not agree [13, Section 3.2]. Their four-dimensional argument uses a more refined function of the Hessian

eigenvalues [13, Section 4].

Our proof uses the normalized shape of the Hessian to construct both a scalar test and a separate vector field whose divergence controls the test's energy. The two constructions are chosen together, so the interior and boundary estimates retain the same tensor data. The proof has three stages.

First, suppose the first nonflat minimizing cone occurs in dimension five or six. Blowups away from its vertex reduce the dimension and are therefore flat. Regularity theory makes each component of its spherical positive set smooth, and at least one component has nonzero Hessian. A five-dimensional cone can be made independent of one extra coordinate without losing minimality. Its spherical link then has precisely two exceptional tips. Thus both possible dimensions lead to a domain $\Omega \subset S^5$ with a smooth regular boundary and, possibly, these two tips (Section 2).

Second, let g be the restriction of the resulting cone to the sphere, and set $T = \nabla^2 g + g \text{Id}$ and $w = |T|$. Where $w > 0$, we construct a positive scalar h from w , the normalized tensor T/w , g , and ∇g , and a vector field Z from the same data and $\nabla T/w$. The target inequalities are

$$\text{div } Z \geq |\nabla h|^2 + 4h^2 + \epsilon_* w^{2a}, \quad N \cdot Z + Hh^2 \geq 0,$$

where $a \in (1/2, 1)$ is fixed, $\epsilon_* > 0$, N points into Ω , and $H = -T(N, N)$. Spherical stability supplies the reverse integrated comparison between the test's energy and its boundary curvature term. The extra positive mass $\epsilon_* w^{2a}$ will give the contradiction. The geometry and stability inequality are developed in Section 3. We then state the universal tensor inequality and derive the two geometric comparisons; the full exact verification follows in Section 6.

Two cancellations make this construction possible. A Laplacian contraction and an antisymmetric curvature contraction eliminate fourth derivatives from $\text{div } Z$. At the boundary, reflection in the normal direction removes the derivative components not determined by the boundary conditions. The remaining inequalities involve only finitely many normalized tensors. We prove them by weighted sums of tensor squares and exact rational remainder bounds, with all coefficients and contraction conventions supplied in Appendix A.

Finally, neither the normalized Hessian nor its derivatives are defined at $w = 0$, and suspension introduces singular tips. These sets cannot be discarded without an estimate. A regularized Bochner identity gives an integrable weight controlling the derivatives near every Hessian zero. The explicit suspension formulas give an integrable power of the distance parameter at both tips. Smooth product cutoffs therefore remove both obstructions and make all error terms vanish. Section 7 completes the stability contradiction.

Organization and conventions. Section 2 reduces a hypothetical first nonflat minimizer to a smooth spherical component in dimension six, allowing two suspension tips. Section 3 derives the tensor identities and the stability inequality. Section 4 states the universal algebraic estimate, and Section 5 derives the pointwise geometric comparison from it. Section 6 gives the complete exact verification; Section 7 then removes the cutoffs to obtain the contradiction. Section 8 gives the regularity and sharpness consequences. Appendix A gives all coefficient data. All tensor norms and contractions use orthonormal frames. Repeated indices are summed, $\Delta = \text{tr } \nabla^2$, and a boundary normal denoted by N points into the positive phase. The curvature $H = -T(N, N)$ used below is not divided by the dimension of the boundary.

2 Reduction to a smooth link and its suspension

We first record the consequences of energy minimality needed to put the argument on a smooth spherical domain. Throughout Section 2, a minimizer has the full ballwise Sobolev comparison

property defined in Section 1.

2.1 The regularity inputs

The ballwise convention also gives comparison on every bounded open set $O \Subset \mathbb{R}^d$: extend an $H_0^1(O)$ competitor difference by zero into a ball containing $\overline{O}$, apply minimality there, and cancel the energy outside O . Conversely, comparison on bounded open sets includes comparison on balls. It is even enough to have comparison against *arbitrary* admissible competitors in every origin-centered ball, by the same extension argument. There is no restriction to homogeneous competitors.

The convention that competitors are nonnegative makes no difference to the standard one-phase results. If $v - u \in H_0^1(O)$ and $u \geq 0$, then $v_+ - u \in H_0^1(O)$ and

$$\int_O (|\nabla v_+|^2 + \mathbf{1}_{\{v_+ > 0\}}) \leq \int_O (|\nabla v|^2 + \mathbf{1}_{\{v > 0\}}).$$

For completeness, the assertion about $H_0^1(O)$ follows by approximating $v - u$ in H^1 by compactly supported smooth functions and applying the continuity of positive-part truncation in H^1 ; the truncated differences in the approximation have compact support in O .

Proposition 2.1 (Regularity and blowups of minimizers). *Let $d \geq 2$ and let $u \geq 0$ be a local minimizer of the one-phase energy with volume coefficient 1 in an open set $D \subset \mathbb{R}^d$.*

- (i) *There is a locally Lipschitz representative of u , and it is harmonic in $\{u > 0\}$.*
- (ii) *If $x_0 \in D \cap \partial\{u > 0\}$, every sequence $r_j \downarrow 0$ has a subsequence for which*

$$u_{x_0, r_j}(y) := \frac{u(x_0 + r_j y)}{r_j} \longrightarrow U(y) \tag{2.1}$$

locally uniformly in $\mathbb{R}^d$. The limit is a nonzero, nonnegative, one-homogeneous global minimizer. Along a further subsequence the convergence is strong in H_{loc}^1 , and the positivity indicators converge in L_{loc}^1 .

- (iii) *If one such limit equals $(y \cdot e)_+$ with $|e| = 1$, then near x_0 the free boundary is a smooth hypersurface and $\{u > 0\}$ is exactly one of its sides. The function u is smooth up to that hypersurface from the positive side, and*

$$u = 0, \quad \nabla u = N_E \quad \text{on the free boundary,} \tag{2.2}$$

where N_E is the Euclidean unit normal pointing into $\{u > 0\}$.

Sources and applicability. We use the constant-coefficient theory with $\Lambda = 1$ in [17]. For each ball $B \Subset D$, the restriction of u to B minimizes against every $H_0^1(B)$ competitor difference. Zero extension of differences therefore gives comparison on every $O \Subset B$, as required by Definitions 2.11–2.12 there. We apply the local results inside such balls. For global minimizers, the containing-ball argument above gives the comparison convention on all bounded open sets. Local Lipschitz continuity is Theorem 3.1. Harmonicity also follows directly: for a smooth test supported compactly in the positive phase, both signs of a sufficiently small additive variation preserve that phase, so the Dirichlet first variation vanishes.

The Lipschitz bound and $u(x_0) = 0$ give locally uniform compactness of the rescalings. Proposition 4.1 gives $\sup_{B_r(x_0)} u \geq \kappa r$ for small r , with $\kappa > 0$, so a limit cannot vanish identically. Proposition 6.2 gives strong Sobolev and positivity-indicator convergence and the full minimizing

property of the limit; Proposition 9.12 gives its one-homogeneity. All these statements can be applied in a bounded ball compactly contained in D before taking the rescalings. In particular, the assertion about minimality is the minimizing compactness result, including the volume term.

Definition 6.10 and Corollary 8.2, with the one-sided graph conclusion in Theorem 8.1, give a $C^{1,\alpha}$ regular neighborhood from a single half-space blowup. Proposition 7.1 supplies the viscosity Bernoulli condition.

To justify all subsequent boundary derivatives, use [6, Theorem 1.1 and Corollary 1.2] with $f_+ = f_- = 0$ and $G(b) = 1 + b$. We check the two-phase viscosity condition of [5, Definition 7.1] when $u \geq 0$. Let v be a comparison touching u at a free-boundary point x_0 , with a C^2 interface and C^2 branches having nonnegative normal slopes a and b on its positive and negative sides. If v touches from below, then $v_+ \leq u$ in a neighborhood. For $a > 0$, extend its positive branch to a signed C^2 function ϕ , negative just across the interface; then $\phi \leq u$ and touches at x_0 . The one-phase condition gives $a \leq 1 \leq G(b)$; the case $a = 0$ is immediate. If v touches from above, the neighborhood inequality $v \geq u \geq 0$ forces $v = u = 0$ throughout its nonpositive side, so $b = 0$. A C^2 extension ϕ of its positive branch satisfies $\phi_+ \geq u$ on both sides and touches at x_0 . Thus the one-phase condition gives $a \geq 1 = G(0)$. These uses of the one-phase condition are justified by [17, Definition 7.6 and Proposition 7.1]: to meet its smooth-test convention, replace each C^2 extension by its quadratic Taylor polynomial minus, respectively plus, $\varepsilon|x - x_0|^2$. For any $\varepsilon > 0$, these are smooth lower and upper supports in a sufficiently small neighborhood, with the same contact value and gradient.

The interior equations hold because u is harmonic where positive and identically zero on the interior of its zero set. The function G is smooth, strictly increasing and unbounded, with $G(0) = 1$; the cited results allow the negative phase to vanish. The preceding $C^{1,\alpha}$ one-sided graph satisfies the slab flatness hypothesis after rotation and sufficiently small dilation, while the local Lipschitz bound is preserved. Theorem 1.1 gives $C^{2,\gamma}$ regularity, and Corollary 1.2 gives smoothness. Local harmonic boundary regularity with zero Dirichlet data [12, Theorems 4.30–4.31 and Corollary 4.32] then gives smoothness of u from the positive side. These estimates apply after flattening a smooth boundary patch and localizing away from its artificial boundary; iteration and Sobolev embedding give every required derivative. The Bernoulli condition and nonnegativity give (2.2). $\square$

As in Section 1, a free-boundary point is regular if it has a half-space blowup and singular otherwise. Proposition 2.1 explains the geometric meaning of regularity. Conversely, a smooth one-sided free boundary with (2.2) has a half-space blowup by its first-order Taylor expansion. We henceforth use the continuous representative of a minimizer. If it is one-homogeneous almost everywhere, it is one-homogeneous everywhere: for each fixed $r > 0$, both sides of $u(rx) = ru(x)$ are continuous and agree almost everywhere.

2.2 Dimension reduction at a nonzero point

The next two lemmas give the dimension-reduction steps explicitly; compare [17, Lemmas 10.9–10.10].

Lemma 2.2 (Translation invariance of an off-origin blowup). *Let u be a one-homogeneous global minimizer, let $0 \neq x_0 \in \partial\{u > 0\}$, and let U be any limit in (2.1). Then*

$$U(y + tx_0) = U(y) \quad (y \in \mathbb{R}^d, t \in \mathbb{R}).$$

Proof. Write $u_r = u_{x_0,r}$. For fixed $t \in \mathbb{R}$ and small $r > 0$, set $r' = r/(1 + rt) > 0$. Homogeneity gives the exact identity

$$u_r(y + tx_0) = \frac{u((1 + rt)(x_0 + r'y))}{r} = u_{r'}(y). \quad (2.3)$$

Let L be a Lipschitz constant for u on a fixed neighborhood of x_0 . Since $u(x_0) = 0$, for $|y| \leq M$ and small r ,

$$\begin{aligned} |u_{r'}(y) - u_r(y)| &\leq \frac{L|r - r'|M}{r'} + LrM \left| \frac{1}{r'} - \frac{1}{r} \right| \\ &= 2LM \frac{|r - r'|}{r'} = 2LM|t|r. \end{aligned}$$

Thus the r' rescalings along the same chosen sequence have the same locally uniform limit as the r rescalings. Passing to that limit in (2.3) proves the assertion for every fixed y, t . No uniqueness of blowups is required. $\square$

Lemma 2.3 (Minimality of a translation-invariant factor). *Suppose $U(y, s) = v(y)$ is a global minimizer in $\mathbb{R}^k \times \mathbb{R}$, where $v \geq 0$ belongs to $H_{\text{loc}}^1(\mathbb{R}^k)$. Then v is a global minimizer in $\mathbb{R}^k$.*

Proof. Suppose that a ball $B \subset \mathbb{R}^k$ admits a nonnegative $q \in H^1(B)$ with $q - v \in H_0^1(B)$ and

$$J(v; B) - J(q; B) = \delta > 0.$$

Let ψ be the zero extension of $q - v$. For $L > 1$, choose $\eta_L : \mathbb{R} \rightarrow [0, 1]$ equal to 1 on $[-L, L]$, equal to 0 outside $[-L - 1, L + 1]$, and linear on each intervening interval. Define

$$V_L(y, s) = U(y, s) + \eta_L(s)\psi(y).$$

This function is nonnegative: on $B \times \mathbb{R}$ it is the convex combination $(1 - \eta_L)v + \eta_L q$. Its difference from U belongs to H^1 and has compact support in a sufficiently large ball $\mathcal{B}_L \subset \mathbb{R}^{k+1}$, so it belongs to $H_0^1(\mathcal{B}_L)$.

The energy difference on $B \times [-L, L]$ is $-2L\delta$. On either transition cylinder, convexity gives

$$|\nabla_y V_L|^2 \leq (1 - \eta_L)|\nabla v|^2 + \eta_L |\nabla q|^2, \quad |\partial_s V_L|^2 \leq |q - v|^2.$$

The difference of the positivity indicators is at most 1. Hence the sum of the two transition contributions is bounded above by

$$C = 2 \int_B (|\nabla q|^2 + |q - v|^2 + 1),$$

independently of L . Outside these three cylinders the energy is unchanged. Thus

$$J(V_L; \mathcal{B}_L) - J(U; \mathcal{B}_L) \leq -2L\delta + C < 0$$

for large L , contradicting minimality of U . $\square$

In dimension one, every nonnegative one-homogeneous Sobolev function has the form $u(x) = b_+ x_+ + b_- (-x)_+$ with $b_+, b_- \geq 0$. If both slopes are positive, a nonnegative smooth bump ϕ supported in $(-R, R)$ with $\phi(0) > 0$ preserves positivity volume and gives

$$J(u + \tau\phi; (-R, R)) - J(u; (-R, R)) = -2\tau(b_+ + b_-)\phi(0) + \tau^2 \int_{-R}^R |\phi'|^2 < 0$$

for small $\tau > 0$. Thus a nonzero minimizing cone has just one positive slope. After reflection write $u(x) = bx_+$, $b > 0$. The competitors $v_t(x) = bR(x - t)_+ / (R - t)$ on $(-R, R)$, $|t| < R$, have the

same trace and energy $b^2 R^2 / (R - t) + (R - t)$. Its derivative at the minimizing parameter $t = 0$ is $b^2 - 1 = 0$, so $b = 1$. This proves flatness in dimension one directly.

The lower-dimensional classification of Jerison–Savin [13, Theorem 1.1 and Corollary 1.2] and the dimension-seven minimizing example of De Silva–Jerison [7, Theorem 1.1] give $5 \leq d_* \leq 7$. De Silva–Jerison’s Theorem 1.1 uses the same functional and arbitrary ballwise Dirichlet competitors. For the exclusion argument, suppose from now on that

$$d = d_* \in \{5, 6\}, \quad (2.4)$$

and fix a nonzero nonflat one-homogeneous global minimizer u in $\mathbb{R}^d$.

Proposition 2.4 (Smooth original link). *Every point of $\partial\{u > 0\} \setminus \{0\}$ is regular. Put $m_o = d - 1$ and $g_o = u|_{S^{m_o}}$. The set $\{g_o > 0\}$ is nonempty and proper, has finitely many connected components, and their closures are pairwise disjoint compact smooth manifolds with boundary. On each such component Ω_o , the restriction of g_o is smooth up to the boundary from within Ω_o and satisfies*

$$\Delta_{S^{m_o}} g_o = -m_o g_o \quad \text{in } \Omega_o, \quad g_o = 0, \quad \nabla_{S^{m_o}} g_o = N_o \quad \text{on } \partial\Omega_o, \quad (2.5)$$

where N_o is the inward spherical unit normal. In particular, $\partial\Omega_o \neq \emptyset$.

Proof. At a nonzero free-boundary point x_0 , choose a blowup U from Proposition 2.1. By Lemma 2.2, in orthogonal coordinates $\mathbb{R}^d = x_0^\perp \times \mathbb{R}$ it has the form $U(y, s) = v(y)$. The factor v is locally Lipschitz, nonnegative, nonzero and one-homogeneous. Lemma 2.3 shows that it is a global minimizer. Since its dimension is $d - 1 < d_*$, it is flat. Therefore U itself is a half-space solution, and Proposition 2.1 makes x_0 regular.

The spherical positive set is nonempty by nontriviality. The polar-coordinate formula for the Euclidean Laplacian of $u(ry) = r g_o(y)$ gives

$$\Delta_{\mathbb{R}^d} u(ry) = r^{-1} (\Delta_{S^{m_o}} g_o(y) + m_o g_o(y)) \quad (2.6)$$

in the positive phase. If $g_o > 0$ on the entire sphere, integrating the harmonicity identity over the sphere would give $m_o \int g_o = 0$, a contradiction. Thus its positive set is proper.

The conical free boundary has the radial direction tangent to it at every nonzero point: the curve $r \mapsto rx$ lies in that free boundary. Consequently its smooth hypersurface is transverse to the unit sphere, and their intersection is a smooth hypersurface of S^{m_o} . Its regular charts have the spherical positive phase exactly on one side. Each point of the sphere therefore has a neighborhood whose intersection with $\{g_o > 0\}$ is either connected or empty (using a small ball for points away from the free boundary). A finite subcover of the sphere shows that there are only finitely many positive components. If two component closures met, their common point would be a free-boundary point; the connected positive side of a regular chart there would meet both components, forcing them to coincide. The same charts show that each closure is a smooth compact domain. Each component has nonempty boundary because the sphere is connected and the positive set is proper.

Smoothness of u from its positive side gives smoothness of g_o on these closures. Equation (2.6) gives the interior equation in (2.5). On the boundary, the radial derivative of u is $g_o = 0$, so the Euclidean gradient is tangent to the sphere. Equation (2.2) then gives the remaining boundary conditions. $\square$

Lemma 2.5 (Selection of a nonflat component). *There is a component Ω_o in Proposition 2.4 such that*

$$T_o := \nabla_{S^{m_o}}^2 g_o + g_o \text{Id} \neq 0 \quad \text{on } \Omega_o. \quad (2.7)$$

Here Id denotes the spherical metric, viewed as a symmetric two-tensor.

Proof. For y in the spherical positive set and $r > 0$,

$$\nabla_{\mathbb{R}^d} u(ry) = g_o(y)y + \nabla_{S^{m_o}} g_o(y).$$

Differentiating this formula, or using Euler's identity for the radial entries, gives

$$D^2 u(ry)[y, \cdot] = 0, \quad D^2 u(ry)[v_1, v_2] = r^{-1} T_o(y)[v_1, v_2] \quad (v_1, v_2 \perp y). \quad (2.8)$$

Suppose that T_o vanished identically on every positive component. Then $D^2 u = 0$ on each corresponding connected open cone D_i , so $u(x) = b_i \cdot x + c_i$ there. Homogeneity forces $c_i = 0$, and positivity forces $b_i \neq 0$. Thus $D_i \subset H_i := \{b_i \cdot x > 0\}$. The set D_i is also closed relative to H_i : if $x \in \overline{D_i} \cap H_i$, continuity gives $u(x) = b_i \cdot x > 0$; a small positive neighborhood of x meets D_i , hence belongs to the same component. Since H_i is connected, $D_i = H_i$. The Bernoulli condition at any nonzero point of its boundary gives $|b_i| = 1$.

Two disjoint open half-spaces through the origin must have opposite unit normals: if their normals are e, f with $f \neq -e$, the vector $e + f$ lies in both. But opposite half-spaces have spherical closures meeting along the equator $e^\perp \cap S^{m_o}$, contrary to the pairwise disjoint closures in Proposition 2.4. Thus there is exactly one positive component, and $u = (x \cdot e)_+$, contrary to nonflatness. This proves (2.7). $\square$

2.3 Suspension and the working domain

Proposition 2.6 (Suspension preserves global minimality). *If $u \geq 0$ is a global minimizer in $\mathbb{R}^k$, then $\tilde{u}(x, s) = u(x)$ is a global minimizer in $\mathbb{R}^{k+1}$. If u is the cone in (2.4) with $d = 5$, the singular free-boundary set of $\tilde{u}$ is exactly $\{0\} \times \mathbb{R}$. For the component chosen in Lemma 2.5, the corresponding positive component on S^5 is*

$$\tilde{\Omega} = \{\Phi(t, y) := (\sin t y, \cos t) : 0 < t < \pi, y \in \Omega_o\}, \quad \tilde{g}(\Phi(t, y)) = \sin t g_o(y). \quad (2.9)$$

Its boundary consists of the smooth side $\Phi((0, \pi) \times \partial\Omega_o)$ and the two singular tips $p_+ = (0, 1)$ and $p_- = (0, -1)$.

Proof. Fix a ball $\mathcal{B} = B_R((x_c, s_c)) \subset \mathbb{R}^{k+1}$ and a nonnegative $V \in H^1(\mathcal{B})$ with $V - \tilde{u} \in H_0^1(\mathcal{B})$. Extend that difference by zero to $F \in H^1(\mathbb{R}^{k+1})$. For $|s - s_c| < R$, let

$$B_s = B_{\sqrt{R^2 - (s - s_c)^2}}(x_c)$$

be the horizontal slice. For almost every such s , $F(\cdot, s)|_{B_s} \in H_0^1(B_s)$. To see the zero-boundary assertion without a trace regularity assumption on V , approximate F in $H^1(\mathbb{R}^{k+1})$ by the zero extensions of functions in $C_c^\infty(\mathcal{B})$. After passing to a subsequence, Fubini's Theorem gives convergence of their slices to $F(\cdot, s)$ in $H^1(\mathbb{R}^k)$ for almost every s . Each approximating slice lies in $C_c^\infty(B_s)$, so the assertion follows by closure.

For almost every s , $V_s = V(\cdot, s)$ is therefore an admissible nonnegative competitor for u in B_s . Integrating its minimizing inequality and using the nonnegativity of the additional derivative energy, we obtain

$$\begin{aligned} J(\tilde{u}; \mathcal{B}) &= \int_{s_c - R}^{s_c + R} J(u; B_s) ds \\ &\leq \int_{s_c - R}^{s_c + R} J(V_s; B_s) ds \leq J(V; \mathcal{B}). \end{aligned}$$

This proves the full ballwise minimizing property, for arbitrary centers and arbitrary admissible Sobolev competitors. One-homogeneity and nontriviality are plainly preserved. Flatness of $\tilde{u}$ would force its half-space normal to have zero last coordinate, and would therefore force u to be flat.

In the conical case, $u(0) = 0$ and a positive ray approaches the origin, so $0 \in \partial\{u > 0\}$. The product identity

$$\partial\{\tilde{u} > 0\} = \partial\{u > 0\} \times \mathbb{R}$$

shows that all its off-axis points are regular: their positive phase is the product of a smooth one-sided regular chart for u and an interval. At any axis point $z_0 = (0, s_0)$, every rescaling is exactly

$$\frac{\tilde{u}(z_0 + r(x, s))}{r} = \frac{u(rx)}{r} = \tilde{u}(x, s).$$

This is nonflat, so no axis point is regular. The axis meets S^5 only in p_+ and p_- .

Away from these two points, Φ is a smooth product coordinate map from $(0, \pi) \times S^4$ to $S^5 \setminus \{p_+, p_-\}$. Positivity in these coordinates is precisely $g_o(y) > 0$, so connected components are as in (2.9). Since $\overline{\Omega_o}$ is compact, its suspended closure is obtained by adding the two endpoints of the interval, with each endpoint collapsed to its tip. Its remaining boundary is exactly the stated smooth side. $\square$

Notation for the remaining argument. Fix the original component $\Omega_o \subset S^{m_o}$ selected in Lemma 2.5, where $m_o \in \{4, 5\}$. If $m_o = 5$, set $\Omega = \Omega_o$, $g = g_o$, and $\Sigma = \emptyset$. If $m_o = 4$, set $\Omega = \tilde{\Omega}$, $g = \tilde{g}$, and $\Sigma = \{p_+, p_-\}$. In either case $\Omega \subset S^5$ is connected, and g comes from a nonnegative one-homogeneous global minimizer in $\mathbb{R}^6$. The closure is smooth away from Σ , with positive phase on one side of its regular boundary

$$\partial_{\text{reg}}\Omega := \partial\Omega \setminus \Sigma.$$

All spherical boundary values below are taken from the positive side. Tests on $\overline{\Omega}$ will be smooth and supported away from Σ . Every compact portion of the selected Euclidean cone's closure that avoids the origin and, in the suspended case, the axis is separated from all other positive components. This follows from the disjoint compact original closures, and in the suspension from the same fact on any closed subinterval of $(0, \pi)$. Consequently the local variations on this component can be made as variations of the global minimizer. No minimizing property of the selected component extended by zero is needed.

3 Spherical geometry and the stability inequality

Let $\Omega_o \subset S^{m_o}$, $m_o \in \{4, 5\}$, be the original component supplied by Proposition 2.4 and Lemma 2.5. Its closure is a compact smooth manifold with nonempty boundary. The function g_o is smooth up to that boundary, is positive in Ω_o , and satisfies

$$\Delta g_o = -m_o g_o, \quad g_o = 0, \quad \nabla g_o = N_o \quad \text{on } \partial\Omega_o,$$

where N_o is the inward unit normal. Moreover, $\nabla^2 g_o + g_o \text{Id}$ does not vanish identically. We give the identities in a general tangent dimension m . The compact original component first provides strict gradient and boundary-sign bounds; we then transport these bounds to the working link and derive its stability inequality. All spherical derivatives below are covariant, and Δ on tensors is the rough Laplacian $\text{tr } \nabla^2$. Repeated indices in an orthonormal frame are summed.

3.1 The spherical Hessian and its boundary values

For a smooth solution of $\Delta g = -mg$ on a domain in S^m , set

$$p = \nabla g, \quad T = \nabla^2 g + g \text{Id}, \quad B_{ijk} = \nabla_i T_{jk}, \quad w = |T|, \quad s^2 = g^2 + |p|^2. \quad (3.1)$$

The notation s^2 denotes the displayed smooth function, including at points where it vanishes. If $u(r y) = r g(y)$ is the homogeneous extension, then in the radial and tangential orthonormal frame

$$D^2 u(r y) = r^{-1} \begin{pmatrix} 0 & 0 \\ 0 & T(y) \end{pmatrix}. \quad (3.2)$$

Indeed, Euler's identity gives $D^2 u y = 0$, and differentiation of $\nabla_{\mathbb{R}^{m+1}} u = g y + p$ along a tangent vector gives the tangential block $\nabla^2 g + g \text{Id}$ at radius one. Homogeneity then gives the factor r^{-1} . The same gradient formula identifies s^2 with $|\nabla_{\mathbb{R}^{m+1}} u|^2$ on the unit sphere.

Proposition 3.1 (Interior identities). *The tensor T is symmetric and trace-free, and B is fully symmetric and trace-free in every pair of slots. In particular $\text{div } T = 0$. The following identities hold:*

$$\nabla_l B_{ijk} - \nabla_i B_{ljk} = \delta_{lj} T_{ik} + \delta_{lk} T_{ij} - \delta_{ij} T_{lk} - \delta_{ik} T_{lj}, \quad (3.3)$$

$$\Delta T = mT, \quad \Delta w^2 = 2mw^2 + 2|B|^2, \quad (3.4)$$

$$\nabla_l p_i = T_{li} - g\delta_{li}, \quad \nabla_l g = p_l, \quad (3.5)$$

$$\nabla s^2 = 2Tp, \quad \Delta s^2 = 2w^2. \quad (3.6)$$

On the open set $\{w > 0\}$ one also has

$$\nabla_l w = w^{-1} T_{jk} B_{ljk}, \quad |\nabla w| \leq |B|. \quad (3.7)$$

The inequality $|\nabla(w^2)|^2 \leq 4w^2|B|^2$ holds everywhere, including at $w = 0$.

Proof. The trace equation is $\text{tr } T = \Delta g + mg = 0$. On the unit sphere the commutation rule for the covector p gives

$$\nabla_i (\nabla^2 g)_{jk} - \nabla_j (\nabla^2 g)_{ik} = \delta_{ik} p_j - \delta_{jk} p_i.$$

Adding the derivatives of $g \text{Id}$ cancels the right side. Thus T is Codazzi: $\nabla_i T_{jk} = \nabla_j T_{ik}$. Together with symmetry in j, k , this proves full symmetry of B . Differentiating $\text{tr } T = 0$ shows $B_{ijj} = 0$; full symmetry gives all other traces and hence $\text{div } T = 0$.

For clarity, the same curvature convention on a covariant two-tensor is

$$([\nabla_l, \nabla_i]T)_{jk} = \delta_{lj} T_{ik} + \delta_{lk} T_{ij} - \delta_{ij} T_{lk} - \delta_{ik} T_{lj}.$$

This proves (3.3). Substitute $(l, i, j, k) = (i, j, i, k)$ there and sum over i . By symmetry and the zero traces of B , the left side is ΔT_{jk} , whereas the right side is

$$mT_{jk} + T_{jk} - T_{jk} - \delta_{jk} \text{tr } T = mT_{jk}.$$

The product rule for $|T|^2$ now proves (3.4). The definitions give (3.5) and

$$\nabla_l s^2 = 2gp_l + 2p_i(T_{li} - g\delta_{li}) = 2T_{li}p_i.$$

Taking the divergence and using $\text{div } T = 0$ and $\text{tr } T = 0$ gives $\Delta s^2 = 2T_{li}(T_{li} - g\delta_{li}) = 2w^2$. Finally, differentiating w^2 and applying Cauchy–Schwarz proves both gradient estimates. $\square$

Lemma 3.2 (Strict gradient bound on the original component). *On Ω_o one has $s_o^2 < 1$. At every boundary point, put*

$$H_o = -T_o(N_o, N_o). \quad (3.8)$$

Then

$$H_o > 0, \quad T_o p_o = -H_o p_o, \quad w_o > 0 \quad \text{on } \partial\Omega_o. \quad (3.9)$$

Consequently w_o has a positive minimum on $\partial\Omega_o$, and its zero set is a compact subset of Ω_o .

Proof. The boundary conditions give $s_o^2 = 1$ on $\partial\Omega_o$. By (3.6), s_o^2 is subharmonic. The maximum principle on the compact closure gives $s_o^2 \leq 1$. If equality held at an interior point, the strong maximum principle would make s_o^2 constant on the connected component, and then $\Delta s_o^2 = 2w_o^2$ would imply $T_o \equiv 0$. Thus $s_o^2 < 1$ inside.

The smooth boundary has the interior sphere property at each point. The Hopf boundary lemma therefore gives $\partial_{N_o} s_o^2 < 0$. Since $p_o = N_o$ there, (3.6) identifies this derivative as $2T_o(N_o, N_o)$, proving $H_o > 0$. For a tangent vector τ to $\partial\Omega_o$, tangential differentiation of $s_o^2 = 1$ similarly gives $T_o(\tau, N_o) = 0$. This proves the eigenvector assertion and $w_o \geq H_o > 0$. Continuity and compactness of the boundary give a positive minimum of w_o and a neighborhood of the boundary disjoint from its zero set. $\square$

The remaining boundary formulas are local. Thus they apply on the original boundary and, once the strict sign has been transported, on the regular side of the suspension.

Proposition 3.3 (Boundary jets). *Suppose $g = 0$, $p = N$, and $T(N, N) = -H < 0$ on a smooth boundary portion, where N is inward. Let Greek indices be tangent to this boundary and put*

$$\mathfrak{h}_{\alpha\beta} = \langle \nabla_\alpha N, e_\beta \rangle.$$

Then $T_{N\alpha} = 0$, $T_{\alpha\beta} = \mathfrak{h}_{\alpha\beta}$, and $H = \text{tr } \mathfrak{h}$; in particular H is a sum of principal curvatures with the specified convention, without division by $m - 1$. If ∇^∂ is the boundary connection, all components of B are described, up to permutation of slots, by

$$B_{\alpha\beta\gamma} = (\nabla_\alpha^\partial \mathfrak{h})_{\beta\gamma}, \quad B_{NN\alpha} = -\partial_\alpha H, \quad (3.10)$$

$$B_{N\alpha\beta} = T_{NN}T_{\alpha\beta} - (T^2)_{\alpha\beta}, \quad B_{NNN} = w^2. \quad (3.11)$$

The components with an odd number of normal slots are thus determined by the pointwise tensor T ; the other components involve tangential derivatives of curvature. The tensor V below encodes this odd-normal part after normalization. Here $w > 0$. Define the normalized tensors and eigenvalue

$$A = T/w, \quad \lambda = T_{NN}/w = -H/w, \quad S_0 = \lambda A - A^2,$$

and the symmetric tensor

$$V_{ijk} = p_i(S_0)_{jk} + p_j(S_0)_{ki} + p_k(S_0)_{ij} + p_i p_j p_k. \quad (3.12)$$

Then $|p| = 1$, $Ap = \lambda p$, $\lambda < 0$, $S_0 p = 0$, $\text{tr } S_0 = -1$, and V is trace-free. The tensors B/w^2 and V agree in every component with an odd number of normal slots. Moreover,

$$V_{Njk} = (S_0 + p \otimes p)_{jk}, \quad \partial_N w = w^2(2\lambda - \text{tr } A^3). \quad (3.13)$$

Proof. On the boundary, tangential differentiation of $|p|^2 = 1$ gives $T_{N\alpha} = 0$, and $\nabla_\alpha p = T e_\alpha$ because $g = 0$. Since $p = N$ along the boundary, this proves $T_{\alpha\beta} = \mathfrak{h}_{\alpha\beta}$. The trace equation for T gives $\text{tr } \mathfrak{h} = -T_{NN} = H$. In differentiating three tangential entries of T , the normal connection terms pair with $T(N, e_\alpha) = 0$. This yields the first formula in (3.10). Likewise,

$$\partial_\alpha T(N, N) = B_{\alpha NN} + 2T(\nabla_\alpha N, N) = B_{\alpha NN},$$

which proves the second formula by symmetry of B .

For a tangent field e_α , differentiate $T(N, e_\alpha) = 0$ along e_β . Since $\langle N, \nabla_\beta e_\alpha \rangle = -\mathfrak{h}_{\beta\alpha}$, this gives

$$0 = B_{\beta N\alpha} + (T^2)_{\beta\alpha} - T_{NN}T_{\beta\alpha}.$$

This proves the first formula in (3.11). Taking its tangential trace gives

$$\sum_\alpha B_{N\alpha\alpha} = T_{NN} \sum_\alpha T_{\alpha\alpha} - \sum_{\alpha,\beta} T_{\alpha\beta}^2 = -T_{NN}^2 - \sum_{\alpha,\beta} T_{\alpha\beta}^2 = -w^2.$$

The zero trace of B then gives $B_{NNN} = w^2$.

Normalization gives $S_0 p = 0$ and $\text{tr } S_0 = -|A|^2 = -1$. Consequently $V_{ijj} = p_i \text{tr } S_0 + 2(S_0 p)_i + p_i = 0$. In an orthonormal frame with first vector $p = N$, one has $V_{N\alpha\beta} = (S_0)_{\alpha\beta}$ and $V_{NNN} = 1$. These are exactly the normalized identities (3.11), and $V_{Njk} = S_{0,jk} + p_j p_k$. Finally, T has only its normal-normal and tangential-tangential blocks, so the odd-normal formulas account for the entire contraction $T_{jk} B_{Njk}$. Thus

$$\partial_N w = w^{-1} T_{jk} B_{Njk} = w^2 A_{jk} (S_0 + p \otimes p)_{jk} = w^2 (2\lambda - \text{tr } A^3),$$

as asserted. $\square$

3.2 Transport to a suspension

Proposition 3.4 (Suspended data). *If $m_o = 4$, write $\rho = \sin t$, $c = \cos t$ and parameterize the regular part of the suspended component by*

$$(t, y) \longmapsto (\rho y, c), \quad 0 < t < \pi, \quad y \in \Omega_o.$$

Its metric is $dt^2 + \rho^2 ds_{S^4}^2$, and $\tilde{g} = \rho g_o$. With $E_0 = \partial_t$ and $E_\alpha = \rho^{-1} e_\alpha$ for a lifted orthonormal frame on S^4 , the data satisfy

$$\tilde{p}_0 = c g_o, \quad \tilde{p}_\alpha = (p_o)_\alpha, \quad \tilde{s}^2 = s_o^2, \quad (3.14)$$

$$\tilde{T}_{00} = 0, \quad \tilde{T}_{0\alpha} = 0, \quad \tilde{T}_{\alpha\beta} = \rho^{-1} (T_o)_{\alpha\beta}. \quad (3.15)$$

The only possibly nonzero components of $\tilde{B}$ are

$$\tilde{B}_{\alpha\beta\gamma} = \rho^{-2} (B_o)_{\alpha\beta\gamma}, \quad \tilde{B}_{0\alpha\beta} = \tilde{B}_{\alpha 0\beta} = \tilde{B}_{\alpha\beta 0} = -c \rho^{-2} (T_o)_{\alpha\beta}. \quad (3.16)$$

In particular,

$$\tilde{w} = \rho^{-1} w_o, \quad |\tilde{B}|^2 = \rho^{-4} (|B_o|^2 + 3c^2 w_o^2), \quad \tilde{H} = \rho^{-1} H_o \quad \text{on the regular side.} \quad (3.17)$$

The volume and side-boundary measures are

$$dV = \rho^4 dt dV_o, \quad d\sigma = \rho^3 dt d\sigma_o. \quad (3.18)$$

Every local identity of Proposition 3.1 holds with $m = 5$, and Proposition 3.3 holds on the regular side. Also $\tilde{s}^2 < 1$ inside and $\tilde{H}, \tilde{w} > 0$ on that side.

Proof. At a point where the base frame is normal, the warped metric has connection

$$\nabla_{E_0} E_0 = \nabla_{E_0} E_\alpha = 0, \quad \nabla_{E_\alpha} E_0 = (c/\rho) E_\alpha, \quad \nabla_{E_\alpha} E_\beta = -(c/\rho) \delta_{\alpha\beta} E_0.$$

The horizontal part of the last expression at other points is the base connection with factor ρ^{-1} . Differentiating $\tilde{g} = \rho g_o$ gives (3.14). Its Hessian has entries

$$(\nabla^2 \tilde{g})_{00} = -\rho g_o, \quad (\nabla^2 \tilde{g})_{0\alpha} = 0, \quad (\nabla^2 \tilde{g})_{\alpha\beta} = \rho^{-1} (\nabla_o^2 g_o)_{\alpha\beta} + (c^2/\rho) g_o \delta_{\alpha\beta}.$$

Adding $\tilde{g} \text{Id}$ proves (3.15). Differentiating that tensor with the displayed connection gives (3.16); entries with two or three zero slots vanish. Each term with exactly one zero slot occurs in three positions, which proves the factor 3 in (3.17).

The inward normal to the side is the horizontal lift of N_o . Thus the normal-normal entry of (3.15) gives $\tilde{H} = \rho^{-1} H_o$. The metric on the side is $dt^2 + \rho^2 ds_{\partial\Omega_o}^2$, proving (3.18). The trace of (3.15) is zero, so $\Delta \tilde{g} = -5\tilde{g}$. The interior identities follow locally from Proposition 3.1. The bounds for s^2 , H and w follow from the original compact component and (3.14)–(3.17); no maximum principle at a tip is used. The local boundary hypotheses and hence Proposition 3.3 hold on the regular side. $\square$

We now use the unadorned notation Ω, g, p, T, B, w, H for the working component in S^5 defined at the end of Section 2. It is either Ω_o when $m_o = 5$, or the above suspension when $m_o = 4$. All boundary integrals mean integrals over the regular boundary, with the two tips omitted in the suspended case. The original data retain the subscript o when needed.

3.3 Second variation under compact perturbations

We derive stability directly from the minimizing property, including its sign and test class. The argument works in either original ambient dimension and in the six-dimensional suspension.

Proposition 3.5 (Euclidean stability on a component). *Let D be the selected positivity component of a global minimizing cone in $\mathbb{R}^{m+1}$. Let ϕ be any real smooth function up to the regular boundary of D , with compact support in that regular closure away from the vertex and from any other singular points. Then*

$$\int_D |\nabla \phi|^2 dx \geq \int_{\partial D} H_E \phi^2 d\sigma_E, \quad H_E = -D^2 u(N_E, N_E), \quad (3.19)$$

where N_E is the inward Euclidean unit normal. No restriction is imposed on the boundary trace or normal derivative of ϕ .

Proof. Every regular free-boundary point has a neighborhood where the positive phase is the single side belonging to D . Compactness of the boundary portion meeting $\text{supp } \phi$ gives finitely many such neighborhoods and a collar separated from every other positivity component and every singular point. Choose the collar over a slightly larger boundary portion, so that the extension of ϕ can vanish near its lateral edge. Extend ϕ smoothly across this boundary with compact support in these neighborhoods. Also extend the positive side of u smoothly to a function F on the collar, with its interior jets unchanged. Since $u = 0$ and $\nabla u = N_E$ on the boundary, shrinking the collar makes F negative on its exterior side and positive on its interior side. Such an extension need only be smooth to finite order sufficient for the expansions below; smooth boundary regularity supplies this, and a partition of unity joins the local extensions while preserving the interior function and its jets.

For a real parameter ε of either sufficiently small sign, use $(F + \varepsilon\phi)_+$ in the collar, $u + \varepsilon\phi$ on the rest of the affected portion of D , and u elsewhere. These definitions agree: $F = u$ on the interior of the collar, and on the compact interior support outside a smaller collar u has a positive minimum.

Near the exterior and lateral edges of the modification the function agrees with u . The resulting competitor u_ε is nonnegative, belongs to H_{loc}^1 , and has compactly supported difference from u in a containing ball. It is therefore admissible for the original global minimizer. This construction uses no separate minimizing assertion for D .

We compute the change in energy in such a ball. First integrate the perturbed density over the fixed original domain D , and then account for the signed boundary layer where its positivity set changes. The fixed-domain contribution is

$$\begin{aligned} 2\varepsilon \int_D \langle \nabla u, \nabla \phi \rangle dx + \varepsilon^2 \int_D |\nabla \phi|^2 dx \\ = -2\varepsilon \int_{\partial D} \phi d\sigma_E + \varepsilon^2 \int_D |\nabla \phi|^2 dx. \end{aligned} \quad (3.20)$$

Here harmonicity eliminates the interior term and the outward normal is $-N_E$, so $\partial_{-N_E} u = -1$. All integrations by parts have compact support on the regular portion.

Use coordinates (y, ξ) obtained by flowing from the original boundary under

$$V_F = \frac{\nabla F}{|\nabla F|^2}, \quad \xi = F.$$

The denominator is nonzero on a sufficiently small compact collar. At $\xi = 0$, $V_F = N_E$. If $j(y, \xi)$ is the volume Jacobian relative to $d\sigma_E(y) d\xi$, then

$$j(y, 0) = 1, \quad \partial_\xi j(y, 0) = \operatorname{div} V_F(y, 0) = \Delta F - 2F_{N_E N_E} = 2H_E. \quad (3.21)$$

The equality $\Delta F = 0$ on the original boundary follows from matching the interior second derivatives; harmonicity outside D is unnecessary. Moreover,

$$\partial_\xi |\nabla F|^2|_{\xi=0} = 2F_{N_E N_E} = -2H_E,$$

and hence

$$\partial_\xi [(1 + |\nabla F|^2)j]|_{\xi=0} = 2H_E. \quad (3.22)$$

Write $\phi_0 = \phi(y, 0)$ and $\phi_N = \partial_{N_E} \phi(y, 0)$. The new zero surface has coordinate $b_\varepsilon(y)$ determined by $b_\varepsilon + \varepsilon \phi(y, b_\varepsilon) = 0$. The implicit function theorem and $\partial_\xi \phi(y, 0) = \phi_N$ give, uniformly on the compact patch,

$$b_\varepsilon = -\varepsilon \phi_0 + \varepsilon^2 \phi_0 \phi_N + O(|\varepsilon|^3). \quad (3.23)$$

Throughout the layer $|\xi| \leq C|\varepsilon|$, its perturbed energy density is

$$(1 + |\nabla(F + \varepsilon \phi)|^2)j = 2 + 2\varepsilon \phi_N + 2H_E \xi + O(\varepsilon^2 + |\varepsilon \xi| + \xi^2). \quad (3.24)$$

Integrating this density from b_ε to 0 is the signed correction, whether the phase expands or contracts. The terms of order at most two are

$$\begin{aligned} \int_{b_\varepsilon}^0 (1 + |\nabla(F + \varepsilon \phi)|^2)j d\xi &= -2b_\varepsilon - 2\varepsilon \phi_N b_\varepsilon - H_E b_\varepsilon^2 + O(|\varepsilon|^3) \\ &= 2\varepsilon \phi_0 + \varepsilon^2 (-2\phi_0 \phi_N + 2\phi_0 \phi_N - H_E \phi_0^2) + O(|\varepsilon|^3) \\ &= 2\varepsilon \phi_0 - \varepsilon^2 H_E \phi_0^2 + O(|\varepsilon|^3). \end{aligned} \quad (3.25)$$

This displays the cancellation of the normal-derivative terms. The uniform remainder is integrable over the compact affected boundary. Adding (3.20) also cancels the linear terms, leaving

$$J(u_\varepsilon) - J(u) = \varepsilon^2 \left(\int_D |\nabla \phi|^2 dx - \int_{\partial D} H_E \phi^2 d\sigma_E \right) + o(\varepsilon^2).$$

The energies here are in a ball containing the modification. Minimality and division by ε^2 prove (3.19). $\square$

The angular constant comes from the sharp radial Hardy threshold, which also enters the cone instability criteria in [13, Propositions 2.1–2.2]. We include the radial calculation to specify the compact supports and the order of limits.

Corollary 3.6 (Spherical stability). *For every real function v smooth up to the regular boundary of the working component $\Omega \subset S^5$, with compact support in its closure away from the tips if present,*

$$\int_{\partial\Omega} H v^2 d\sigma \leq \int_{\Omega} (|\nabla v|^2 + 4v^2) dV. \quad (3.26)$$

The trace of v on the regular boundary is arbitrary.

Proof. More generally, consider a link component $\Omega \subset S^m$ with angular tests supported in its regular closure, and write $D = \{ry : r > 0, y \in \Omega\}$. On its regular side the normal is tangent to S^m , and (3.2) gives $H_E(ry) = r^{-1}H(y)$. Polar volume has factor $r^m dr$ and side area has factor $r^{m-1} dr$. For $\zeta \in C_c^\infty((0, \infty))$ and an allowed angular test v , put $\phi(ry) = \zeta(r)v(y)$ in Proposition 3.5. It gives

$$\begin{aligned} \left(\int_0^\infty r^{m-2} \zeta^2 dr \right) \int_{\partial\Omega} H v^2 d\sigma &\leq \left(\int_0^\infty r^{m-2} \zeta^2 dr \right) \int_{\Omega} |\nabla v|^2 dV \\ &\quad + \left(\int_0^\infty r^m |\zeta'|^2 dr \right) \int_{\Omega} v^2 dV. \end{aligned}$$

For each fixed ζ and v , the Euclidean support lies in a finite annulus and is separated from the singular set and other components, so the preceding second variation applies.

Set $\beta = (m-1)/2$. For $f_\zeta(r) = r^\beta \zeta(r)$, integration of the cross term gives the radial Hardy identity

$$\int_0^\infty r^m |\zeta'|^2 dr - \beta^2 \int_0^\infty r^{m-2} \zeta^2 dr = \int_0^\infty r |f'_\zeta(r)|^2 dr \geq 0.$$

To approach its constant, choose nonzero $\psi \in C_c^\infty(\mathbb{R})$ and set $\zeta_k(r) = r^{-\beta} \psi((\log r)/k)$. The substitution $\tau = (\log r)/k$ yields

$$\frac{\int_0^\infty r^m |\zeta'_k|^2 dr}{\int_0^\infty r^{m-2} \zeta_k^2 dr} = \beta^2 + \frac{1}{k^2} \frac{\int_{\mathbb{R}} |\psi'|^2 d\tau}{\int_{\mathbb{R}} \psi^2 d\tau}. \quad (3.27)$$

Indeed the numerator before division is $k \int_{\mathbb{R}} (-\beta \psi + k^{-1} \psi')^2 d\tau$, and $\int \psi \psi' = 0$ by compact support. Divide the separated stability inequality by its positive radial factor and let $k \rightarrow \infty$. In tangent dimension $m = 5$, one has $\beta^2 = 4$, proving (3.26). The same computation on an original link with $m_o = 4$ would give $9/4$; the working link always has $m = 5$.

The order of limits is fixed: for each angular test and each finite k , first take $\varepsilon \rightarrow 0$ in the second variation, and then take $k \rightarrow \infty$. Angular cutoffs used later are removed only after (3.26) has been obtained for each fixed smooth test. The collar and the allowable size of ε may depend on these fixed supports. $\square$

4 A universal algebraic certificate

The comparison needed for the stability inequality is a statement about finite-dimensional tensors. We state it with free algebraic data before deriving its geometric consequences. All tensor spaces here are over $\mathbb{R}^5$, with the Euclidean inner product and full tensor norm; repeated indices range from 1 to 5 and are summed. A symmetric tensor is called trace-free if contraction of any two slots is zero. We write $\text{STF}^r(\mathbb{R}^5)$ for the space of symmetric trace-free tensors of rank r . Fix

$$a = \frac{29}{50}, \quad Q = 250000, \quad \epsilon_* = \frac{438907}{12000000}, \quad \delta_* = \frac{6984137359}{31250000000}. \quad (4.1)$$

Let A be a symmetric trace-free matrix with $|A|^2 = \text{tr } A^2 = 1$, and let X be a symmetric trace-free three-tensor. Set

$$q_l = X_{lij}A_{ij}, \quad \chi = 1 - |p|^2 - g^2,$$

where $p, \ell \in \mathbb{R}^5$ and $g, w, z \in \mathbb{R}$ are otherwise free. In particular, w and z need not be positive. On tensor polynomials in A, p, g, δ , define D_l by the product rule and

$$D_l A_{ij} = X_{lij} - q_l A_{ij}, \quad D_l p_i = w A_{li} - g z \delta_{li}, \quad D_l g = z p_l, \quad D_l \delta_{ij} = 0. \quad (4.2)$$

This is a first derivative with values in the enlarged polynomial algebra containing X, w, z ; no derivative of these new variables is used.

Proposition 4.1 (Universal inequalities). *There are rational tensor polynomials $f, U_{jk}, K_{lijk}, M_l, P_l$ in A, p, g, δ , with the following properties. They are equivariant under orthogonal changes of coordinates, with f invariant. Their coefficients are specified by Table I in Appendix A, using the decoding in Section 6.1. Put*

$$C_{lijk} = K_{lijk} - K_{iljk}, \quad G_{lijk} = \delta_{li} U_{jk} + C_{lijk}.$$

For every choice of the data above with $\chi \geq 0$,

$$\begin{aligned} \mathcal{I} := & z^2 [5U_{jk}A_{jk} + C_{lijk}(\delta_{lj}A_{ik} + \delta_{lk}A_{ij}) - 4f] \\ & + [D_l G_{lijk} + (2a - 1)q_l G_{lijk}] X_{ijk} \\ & + w [D_l M_l + (2a + 1)q_l M_l] + z [D_l P_l + 2aq_l P_l] \\ & + f|\ell|^2 - (D_l f + 2aq_l f)\ell_l \geq \epsilon_*(z^2 + |\ell|^2). \end{aligned} \quad (4.3)$$

In particular, $f \geq \epsilon_*$ whenever $g^2 + |p|^2 \leq 1$.

For every symmetric trace-free unit matrix A and vector p satisfying

$$g = 0, \quad |p| = 1, \quad Ap = \lambda p, \quad \lambda < 0,$$

define

$$S_0 = \lambda A - A^2, \quad V_{ijk} = p_i(S_0)_{jk} + p_j(S_0)_{ki} + p_k(S_0)_{ij} + p_i p_j p_k.$$

Then

$$\mathcal{S} := -\lambda f + U_{jk}(S_0 + p \otimes p)_{jk} + C_{lijk} p_l V_{ijk} + p_l M_l \geq \delta_*. \quad (4.4)$$

Moreover, U and K are even in p , and $P = 0$ when $g = 0$.

We prove Proposition 4.1 in Section 6. First, Section 5 derives its geometric consequences using only the stated properties. The calculation explains the Laplacian and skew contractions in G , the three coefficients of q , and the free vector ℓ .

5 From the algebraic certificate to a geometric flux

We work on the regular part of the spherical component $\Omega \subset S^5$ constructed in Section 2, and put

$$\Omega_+ = \{x \in \Omega : w(x) > 0\}, \quad a = \frac{29}{50}.$$

All assertions up to the side boundary use derivatives from the positive side. In the suspended case the two tips are excluded throughout Section 5. We apply the universal inequalities of Proposition 4.1, whose proof follows in Section 6. For its coefficient polynomials f, U, K, M, P , with their stated slot orders, set

$$C_{lijk} = K_{lijk} - K_{iljk}, \quad G_{lijk} = \delta_{li}U_{jk} + C_{lijk}. \quad (5.1)$$

Thus $C_{iljk} = -C_{lijk}$. No other slot symmetry of U or K will be needed. The orthogonal equivariance asserted in Proposition 4.1 ensures that all the fields defined below are independent of the orthonormal frame.

5.1 Normalized differentiation and the fourth derivatives

On Ω_+ define

$$A = \frac{T}{w}, \quad X = \frac{B}{w}, \quad q_l = X_{lj}A_{jk}. \quad (5.2)$$

The identities of Proposition 3.1 show that $A \in \text{STF}^2(\mathbb{R}^5)$, $|A| = 1$, and $X \in \text{STF}^3(\mathbb{R}^5)$. Moreover, $g^2 + |p|^2 \leq 1$, so these are admissible data for the interior certificate. The homogenizing variable in that certificate is set to $z = 1$ in every geometric application.

Lemma 5.1. *At the data (5.2), the derivation D_l of (4.2), evaluated at $z = 1$, equals covariant differentiation on every coefficient polynomial in A, p, g, δ . In particular,*

$$\nabla_l w = wq_l, \quad \nabla_l A_{ij} = X_{lij} - q_l A_{ij}, \quad \nabla_l p_i = wA_{li} - g\delta_{li}, \quad \nabla_l g = p_l. \quad (5.3)$$

Furthermore,

$$\frac{1}{w}G_{lijk}\nabla_l B_{ijk} = 5U_{jk}A_{jk} + C_{lijk}(\delta_{lj}A_{ik} + \delta_{lk}A_{ij}). \quad (5.4)$$

Proof. Differentiating $w^2 = T_{jk}T_{jk}$ gives $\nabla_l w = T_{jk}B_{lj}/w = wq_l$. The quotient rule then gives the formula for ∇A , and $T = \nabla^2 g + g \text{Id}$ gives the formula for ∇p . In a covariantly constant orthonormal frame at the point under consideration these are exactly the generator rules for D_l . The product rule and compatibility of the connection with contractions prove the assertion for every coefficient polynomial. The rules also preserve the normalization constraints:

$$D_l(\text{tr } A) = 0, \quad D_l(|A|^2) = 2q_l - 2q_l = 0$$

on the admissible data.

Here D_l is applied only to coefficient polynomials in A, p, g, δ . Differentiation of the factors w and X , which are not coefficient variables, is performed separately. In particular, no rule for $D_l X$ or $D_l q$ is being assumed.

For the U part of (5.4),

$$\frac{1}{w}\delta_{li}U_{jk}\nabla_l B_{ijk} = \frac{1}{w}U_{jk}\Delta T_{jk} = 5U_{jk}A_{jk}.$$

For the skew part, the curvature identity in Proposition 3.1 yields

$$\begin{aligned}
\frac{1}{w}C_{lijk}\nabla_l B_{ijk} &= \frac{1}{2w}C_{lijk}(\nabla_l B_{ijk} - \nabla_i B_{ljk}) \\
&= \frac{1}{2}C_{lijk}(\delta_{lj}A_{ik} + \delta_{lk}A_{ij} - \delta_{ij}A_{lk} - \delta_{ik}A_{lj}) \\
&= C_{lijk}(\delta_{lj}A_{ik} + \delta_{lk}A_{ij}).
\end{aligned}$$

The last equality follows by interchanging the summed indices l, i in the last two terms and using $C_{ilk} = -C_{lik}$. Equivalently, $C_{lijk}\nabla_l B_{ijk}$ equals $K_{lijk}(\nabla_l B_{ijk} - \nabla_i B_{ljk})$; thus the factor $1/2$ belongs to the formula with C , not to the formula with K .

This calculation eliminates every undetermined fourth derivative of g . Indeed, if two possible tensors $\nabla_l B_{ijk}$ have the same Laplacian contraction and the same skew difference in l, i , their difference has zero contraction with $\delta_{li}U_{jk}$ and with C_{lijk} . No constraint on the remaining fourth derivative entries is required. $\square$

5.2 The square root and the interior comparison

Proposition 4.1 gives

$$f(A, p, g) \geq \epsilon_* = \frac{438907}{12000000} > 0 \quad (5.5)$$

on the entire normalized coefficient domain. The verification in Section 6.5 obtains this bound using the free algebraic data $X = w = z = 0$. Since f is independent of w , this permitted algebraic substitution does not impose $w = 0$ on the geometric data: the fields below are defined on Ω_+ , where $w > 0$.

Define on Ω_+

$$h = w^a \sqrt{f}, \quad Z_l = w^{2a}(G_{lijk}X_{ijk} + wM_l + P_l). \quad (5.6)$$

These fields are smooth on Ω_+ and extend smoothly to each regular side boundary point, since $w > 0$ there by Proposition 3.3.

Proposition 5.2. *The scalar h and vector field Z satisfy*

$$\operatorname{div} Z \geq 4h^2 + |\nabla h|^2 + \epsilon_* w^{2a} \quad \text{on } \Omega_+. \quad (5.7)$$

Proof. First,

$$\nabla_l X_{ijk} = \frac{1}{w} \nabla_l B_{ijk} - q_l X_{ijk}.$$

Applying the product rule to each of the three summands of Z , and then using (5.4), gives the exact identity

$$\begin{aligned}
w^{-2a} \operatorname{div} Z &= 5U_{jk}A_{jk} + C_{lijk}(\delta_{lj}A_{ik} + \delta_{lk}A_{ij}) \\
&\quad + [D_l G_{lijk} + (2a - 1)q_l G_{lijk}]X_{ijk} \\
&\quad + w[D_l M_l + (2a + 1)q_l M_l] + D_l P_l + 2aq_l P_l.
\end{aligned} \quad (5.8)$$

Every D_l in Equation (5.8) is evaluated at $z = 1$. The three coefficients of q come respectively from differentiating $w^{2a}X$, w^{2a+1} , and w^{2a} ; in the first expression the quotient $X = B/w$ contributes $-q_l X$. Thus their values are exactly

$$2a - 1 = \frac{4}{25}, \quad 2a + 1 = \frac{54}{25}, \quad 2a = \frac{29}{25}.$$

Put

$$\mathcal{F}_l = D_l f + 2aq_l f, \quad \ell_* = \frac{\mathcal{F}}{2f}.$$

Differentiating h and completing the square give, respectively,

$$\nabla_l h = \frac{w^a}{2\sqrt{f}} \mathcal{F}_l, \quad |\nabla h|^2 = \frac{w^{2a}}{4f} |\mathcal{F}|^2, \quad (5.9)$$

and

$$f|\ell|^2 - \mathcal{F} \cdot \ell = f|\ell - \ell_*|^2 - \frac{|\mathcal{F}|^2}{4f}.$$

In particular, substituting (5.8) into the interior polynomial $\mathcal{I}$ of (4.3) gives the identity, for every ℓ ,

$$\mathcal{I}|_{z=1} = w^{-2a} (\operatorname{div} Z - 4h^2 - |\nabla h|^2) + f|\ell - \ell_*|^2. \quad (5.10)$$

The division by f is justified uniformly by (5.5). Evaluating (4.3) at $\ell = \ell_*$ therefore gives the slightly stronger estimate

$$\operatorname{div} Z - 4h^2 - |\nabla h|^2 \geq \epsilon_* w^{2a} (1 + |\ell_*|^2),$$

which implies (5.7). □

5.3 Boundary reflection and the signed flux

At a regular side boundary point, Proposition 3.3 gives

$$g = 0, \quad p = N, \quad |p| = 1, \quad Ap = \lambda p, \quad \lambda = -\frac{H}{w} < 0.$$

As in the boundary certificate, define

$$S_0 = \lambda A - A^2, \quad V_{ijk} = p_i(S_0)_{jk} + p_j(S_0)_{ki} + p_k(S_0)_{ij} + p_i p_j p_k. \quad (5.11)$$

Lemma 5.3. *At such a point,*

$$p_l G_{lijk} \frac{B_{ijk}}{w^2} = p_l G_{lijk} V_{ijk} = U_{jk}(S_0 + p \otimes p)_{jk} + C_{lijk} p_l V_{ijk}. \quad (5.12)$$

Proof. Let $R = \operatorname{Id} - 2p \otimes p$ be reflection in $p^\perp$. Since A is symmetric and $Ap = \lambda p$, it preserves $p^\perp$, and hence $RAR^T = A$. Also $Rp = -p$. Orthogonal equivariance and the even parity in p of U, K , and therefore of G , give

$$R^{\otimes 4} G(A, p, 0) = G(RAR^T, Rp, 0) = G(A, -p, 0) = G(A, p, 0).$$

For the three-tensor $F_{ijk} = p_l G_{lijk}$, contraction commutes with orthogonal transformations, so

$$R^{\otimes 3} F = (Rp)_l (R^{\otimes 4} G)_{lijk} = -F.$$

Choose any orthonormal basis with its last vector p . A component of a three-tensor with k normal slots is multiplied by $(-1)^k$ under $R^{\otimes 3}$. It follows that F has zero components whenever the number of normal slots is even. This assertion concerns the full unsymmetrized tensor F ; no permutation of its slots, and no extra symmetry of U or K , has been used.

For tangent indices α, β , the boundary jet identities give

$$\frac{B_{N\alpha\beta}}{w^2} = \lambda A_{\alpha\beta} - (A^2)_{\alpha\beta} = (S_0)_{\alpha\beta}, \quad \frac{B_{NNN}}{w^2} = 1.$$

Because $S_0 p = 0$, these are precisely the components of V with one or three normal slots, including all their permutations. Thus the part of B/w^2 with an odd number of normal slots equals V ; V itself has no even-normal components. The even-normal components of B , which are not determined by A, p, g, w , have zero contraction with F . This proves the first equality in (5.12).

Finally, $S_0 p = 0$ and $|p| = 1$ imply

$$p_i V_{ijk} = (S_0)_{jk} + p_j p_k.$$

Splitting G according to (5.1) proves the second equality. The argument used only the orthogonal splitting $\mathbb{R}^5 = p^\perp \oplus \mathbb{R}p$. It therefore applies without any assumption on multiplicities, signs, or nonvanishing of the tangential eigenvalues of A . $\square$

Proposition 5.4. *On the regular side boundary,*

$$N \cdot Z + Hh^2 = w^{2a+1} \mathcal{S} \geq \delta_* w^{2a+1} > 0, \quad \delta_* = \frac{6984137359}{31250000000}, \quad (5.13)$$

where $\mathcal{S}$ is the boundary polynomial in (4.4). In particular $N \cdot Z + Hh^2 \geq 0$.

Proof. Proposition 4.1 gives $P = 0$ at the boundary, where $g = 0$. Since $X = B/w$, Lemma 5.3 gives

$$\begin{aligned} N \cdot Z &= w^{2a+1} (U_{jk}(S_0 + p \otimes p)_{jk} + C_{lijk} p_l V_{ijk} + p_l M_l), \\ Hh^2 &= -\lambda w^{2a+1} f. \end{aligned}$$

Their sum is exactly $w^{2a+1} \mathcal{S}$. The boundary data satisfy all hypotheses of (4.4), proving the stated lower bound. The sign is expressed using the inward normal N ; the outward normal used by the divergence theorem is $-N$.

Only the boundary value of P vanishes. Its derivative term $D_l P_l$ remains in (5.8). For example, a monomial $P_l = g Q_l$ has $D_i P_l = p_i Q_l + g D_i Q_l$ at $z = 1$, so its derivative need not vanish when $g = 0$. $\square$

5.4 Uniform estimates for the cutoff argument

Lemma 5.5. *There is a finite constant C , depending only on the coefficient polynomials, the dimension, a , and ϵ_* , such that on Ω_+*

$$|h| \leq Cw^a, \quad |\nabla h| \leq Cw^a(1 + w + |X|), \quad |Z| \leq Cw^{2a}(1 + w + |X|). \quad (5.14)$$

The constant is independent of w , $|X|$, and all subsequent cutoff parameters.

Proof. The coefficient domain

$$\{(A, p, g) : A \in \text{STF}^2(\mathbb{R}^5), |A| = 1, |p|^2 + g^2 \leq 1\}$$

is compact. The coefficient polynomials and their first derivatives with respect to their coefficient variables are therefore uniformly bounded there. Also $|q| \leq |X|$ by Cauchy–Schwarz. The rules (5.3) consequently imply

$$|Df| + |q| |f| \leq C(1 + w + |X|).$$

Use (5.5) and (5.9) for the estimates on h and ∇h , and use (5.6) and the boundedness of G, M, P for the estimate on Z . $\square$

6 Exact verification of the algebraic certificate

We return to the free algebraic data of Section 4 to prove Proposition 4.1. All tensors are again over $\mathbb{R}^5$. The scalars w, z and vector ℓ are free, and X is an arbitrary symmetric trace-free three-tensor. Table I specifies the polynomials f, U, K, M, P , and Table II specifies nonnegative comparison forms. For the interior inequality, we compare $\mathcal{I}$ with a positive diagonal quadratic form in (w, z, ℓ, X) plus projected tensor squares, and bound the remaining error term by term. For the boundary inequality, the analogous comparison uses scalar squares in the four tangential eigenvalues and a uniform bound on eleven remainder coefficients. We give the contraction rules and rational bounds in full.

6.1 Tensor monomials and the fixed polynomials

Here and in Appendix A, a rank- r tensor monomial is encoded by a word of decimal digits, partitioned into triples. Its first triple hbj specifies

$$g^h (\text{tr } A^3)^b w^{\mathbf{1}_{\{j=1\}}} z^{\mathbf{1}_{\{j=2\}}}.$$

Each later triple uvk specifies an edge carrying the matrix A^k , with $A^0 = \delta$. An endpoint labeled $i \in \{0, \dots, r-1\}$ is the i th free slot, starting with slot zero. An endpoint labeled 5 carries one copy of p , and one labeled 6 carries one copy of ℓ . Three incidences of label 7 are contracted against the three slots of one X ; three incidences of label 8, if present, use a second copy of X . Each occurrence of a vector endpoint is a separate vector factor. Matrix entries, vector entries, and tensor entries are multiplied, and all nonfree slots are summed. Edges are unoriented because A is symmetric. Permuting the three incidences at an X does not change its value because X is symmetric. There are no implicit multiplicities or factorials in this convention.

Write $\mathcal{M}_r(s)$ for the tensor encoded by the word s . For example,

$$\mathcal{M}_0(200550) = g^2 |p|^2, \quad \mathcal{M}_4(000500520131)_{lijk} = p_l p_j A_{ik}, \quad \mathcal{M}_1(000771700)_l = q_l.$$

The digit labels encode slots and must be distinguished from the coordinate indices that are summed from 1 to 5.

6.1.1 Table I

For $F \in \{f, U, K, M, P\}$, let r_F be respectively 0, 2, 4, 1, 1, with slot orders respectively empty, (j, k) , (l, i, j, k) , (l) , (l) . If a row of Table I consists of F, s, c , its contribution is $c \mathcal{M}_{r_F}(s)/Q$; thus each polynomial is given by

$$F = \frac{1}{Q} \sum_{(F,s,c) \text{ in Table I}} c \mathcal{M}_{r_F}(s). \quad (6.1)$$

Table I contains 34, 57, 34, 74, 32 rows for f, U, K, M, P , respectively. Every free slot occurs exactly once in every row, and these rows have no X, ℓ, w, z factors. Every P row has a positive g exponent. Because an edge has two endpoints and U, K have even rank, their numbers of p endpoints are even. This proves the parity assertions of Proposition 4.1 directly from the definitions. The same definitions show that all five polynomials are equivariant under orthogonal changes of coordinates, with f invariant.

6.2 The nonnegative comparison polynomials

The positive tensor squares below have the form of Gram-matrix certificates for polynomial positivity; compare [15, Section 3.2]. Here all rational polynomials, projection identities, and remainder bounds are explicit, and we verify the certificate directly.

A header @ r kind mode in Table II begins a new block; consecutive equal headers still designate different blocks. For a block of nonnegative rank, write its monomial rows as $s_1, \dots, s_N$ and its rectangular integer coefficient matrix as (c_{tc}) . Define

$$J_c = \frac{1}{Q} \sum_{t=1}^N c_{tc} \mathcal{M}_r(s_t).$$

Its contribution to Γ is

$$\rho \sum_c \langle J_c, \Pi J_c \rangle, \quad \rho = \begin{cases} 1, & \text{mode} = \text{I}, \\ \chi, & \text{mode} = \text{B}. \end{cases} \quad (6.2)$$

Here Π is specified below by r and kind. There are nineteen such blocks with a total of 529 monomial rows. Every row has degree one in (w, z, ℓ, X) , assigning degree one to each component of these variables and degree zero to A, p, g . Hence Γ has degree two in these variables.

The two remaining blocks have rank -1 and fourteen rows each. In these blocks the four digits of a row word are an exponent $\beta = (\beta_1, \dots, \beta_4)$, encoding x^β for $x \in \mathbb{R}^4$. Both coefficient matrices have five columns. Set

$$J_c(x) = \frac{1}{Q} \sum_t c_{tc} x^{\beta_t}, \quad \text{Av } F(x) = \frac{1}{24} \sum_{\pi \in S_4} F(x_{\pi(1)}, \dots, x_{\pi(4)}).$$

Their sum defines the scalar polynomial

$$\Lambda(x) = \sum_{\text{rank } -1 \text{ blocks}} \text{Av} \left(\rho(x) \sum_c J_c(x)^2 \right), \quad \rho(x) = \begin{cases} 1, & \text{mode} = \text{I}, \\ x_1 + x_2 + x_3 + x_4, & \text{mode} = \text{H}. \end{cases} \quad (6.3)$$

Consequently $\Lambda \geq 0$ when $\sum x_i \geq 0$.

6.2.1 Scaled projection operators

For an arbitrary rank- r tensor F , let F^π denote slot permutation by π . We use the averaged symmetrization $\text{Sym } F = (r!)^{-1} \sum_{\pi \in S_r} F^\pi$. The operator Tr_{ab} contracts slots a, b with the Euclidean metric, leaving the remaining slots in their original order. The label **none** means $\Pi = \text{Id}$. The label **anti** means

$$\Pi F = \sum_{\pi \in S_r} \text{sgn}(\pi) F^\pi.$$

For **sym**, put $F' = \sum_{\pi \in S_r} F^\pi$ and $v = \text{Tr}_{12} F'$. The operators used in Table II are

$$(\Pi F')_{ij} = 5F'_{ij} - \delta_{ij}v \quad (r=2), \quad (6.4)$$

$$(\Pi F)_{ijk} = 7F'_{ijk} - (\delta_{ij}v_k + \delta_{ik}v_j + \delta_{jk}v_i) \quad (r=3). \quad (6.5)$$

For $r=3$ and **hook**, set

$$\begin{aligned} F' &= 3(F + F^{(12)}) - \sum_{\pi \in S_3} F^\pi, & v &= \text{Tr}_{12} F', \\ (\Pi F)_{ijk} &= 8F'_{ijk} - (2\delta_{ij}v_k - \delta_{ik}v_j - \delta_{jk}v_i). \end{aligned} \quad (6.6)$$

Lemma 6.1. *All these operators are self-adjoint and positive semidefinite on their entire tensor spaces. In particular, $\Gamma \geq 0$ whenever $\chi \geq 0$.*

Proof. Each slot permutation is orthogonal, with adjoint its inverse. Averaging over all permutations, with or without their signs, gives the orthogonal projection onto symmetric or alternating tensors. Thus **anti** is $r!$ times an orthogonal projection. For a symmetric matrix S , $S - \delta \operatorname{tr} S/5$ is its orthogonal trace-free projection. Since $F' = 2 \operatorname{Sym} F$ in (6.4), that operator is ten times the symmetric trace-free projection.

For rank three define

$$(\mathcal{J}v)_{ijk} = \delta_{ij}v_k + \delta_{ik}v_j + \delta_{jk}v_i.$$

Then $\operatorname{Tr}_{12} \mathcal{J}v = 7v$, and for symmetric S , $\langle S, \mathcal{J}v \rangle = 3\langle \operatorname{Tr}_{12} S, v \rangle$. It follows that $S - \mathcal{J}(\operatorname{Tr}_{12} S)/7$ is the orthogonal projection onto symmetric trace-free tensors. Since now $F' = 6 \operatorname{Sym} F$, (6.5) is 42 times that projection.

For completeness, the trace correction in (6.6) can be checked without any representation-theoretic assumptions. Let

$$S_{12} = \frac{\operatorname{Id} + (12)}{2}, \quad S = \frac{1}{6} \sum_{\pi \in S_3} \pi, \quad P_0 = S_{12} - S.$$

The identities $S_{12}S = SS_{12} = S$ show that P_0 is an orthogonal projection. Its image $\mathcal{W}$ consists of first-pair symmetric tensors orthogonal to fully symmetric tensors. For $H \in \mathcal{W}$,

$$H_{ijk} = H_{jik}, \quad H_{ijk} + H_{ikj} + H_{jki} = 0.$$

If $v = \operatorname{Tr}_{12} H$, contraction of the second identity gives $\operatorname{Tr}_{13} H = \operatorname{Tr}_{23} H = -v/2$. Define

$$(\mathcal{B}v)_{ijk} = 2\delta_{ij}v_k - \delta_{ik}v_j - \delta_{jk}v_i.$$

Direct substitution gives $\mathcal{B}v \in \mathcal{W}$, $\operatorname{Tr}_{12} \mathcal{B}v = 8v$, and

$$\langle H, \mathcal{B}z \rangle = 2\langle v, z \rangle - \langle \operatorname{Tr}_{13} H, z \rangle - \langle \operatorname{Tr}_{23} H, z \rangle = 3\langle v, z \rangle.$$

Therefore $Q_0H = H - \mathcal{B}(\operatorname{Tr}_{12} H)/8$ has all traces zero, and $H - Q_0H$ is orthogonal to every trace-free tensor in $\mathcal{W}$. Thus Q_0 is the orthogonal projection in $\mathcal{W}$ onto its trace-free part. Since $F' = 6P_0F$, the hook operator is $48Q_0P_0$; Q_0P_0 is the orthogonal projection in the full tensor space onto that subspace. In particular

$$\langle F, \Pi F \rangle = 48|Q_0P_0F|^2 \geq 0.$$

Finally the weights in (6.2) are nonnegative under the stated constraint. $\square$

6.3 Exact contraction and differentiation rules

We give explicit collection rules both to specify the finite calculation and to justify that it is an identity on all the constrained tensors. A graph key records the rank, powers of g, w, z , a multiset of trace factors $\operatorname{tr} A^k$, and a multiset of edges with their powers. For purposes of ordering keys, replace endpoint labels 5, 6, 7, 8 by $-1, -2, -3, -4$. Sort the endpoints of each edge, sort the edge list and trace list, and take the lexicographically larger edge list obtained by retaining or exchanging $-3, -4$. A lone cubic tensor is thereby labeled -3 . These choices are only ordering conventions: exchange of two identical copies of X preserves the value of every graph.

Tensor multiplication takes the disjoint union of graphs, shifts the free slot labels of the second factor, and distinguishes any two cubic tensor copies. A contraction of two free slots joins their

incident edges. If these are different edges, their powers add; if they are the same edge, it becomes a trace $\text{tr } A^k$. The removed free slots are deleted and the others renumbered in order. The identities underlying these operations are precisely

$$\sum_i (A^h)_{ai} (A^k)_{ib} = (A^{h+k})_{ab}, \quad \sum_i (A^k)_{ii} = \text{tr } A^k.$$

Reduce the trace factors of powers 0, 1, 2 to 5, 0, 1, respectively. An edge of power zero connecting two slots of the same X makes the monomial zero. Thus every reduction is an equality on the allowed data. Other identities need not be imposed: retaining two different keys for functions that happen to agree cannot invalidate either the expansion or the later triangle inequality.

The derivative of a path and of a trace is

$$D_l(A^k) = \sum_{h=0}^{k-1} A^h X_l A^{k-1-h} - k q_l A^k, \quad (X_l)_{ij} = X_{lij}, \quad (6.7)$$

$$D_l \text{tr } A^k = k \text{tr}(A^{k-1} X_l) - k q_l \text{tr } A^k. \quad (6.8)$$

For $k = 0$ the sums and derivatives are zero. These formulas follow by differentiating each matrix factor using (4.2); cyclic invariance of trace makes all k insertion positions on a trace equal. In graph form, inserting X_l on an edge of power k splits it into two paths of lengths $h, k - 1 - h$ ending at a new X , whose last slot is the derivative index l . The negative terms from all A factors combine into minus the total A degree times q_l times the original graph. Each p endpoint is in turn replaced by $w A_l - g z \delta_l$, and a factor g^h gives $h z g^{h-1} p_l$. If an edge has two p endpoints, these are two product-rule terms, including their multiplicity when they coincide.

These operations are also sound if traces were simplified first, since

$$D_l \text{tr } A = 0, \quad D_l \text{tr } A^2 = 2 X_{lij} A_{ij} - 2 q_l |A|^2 = 0.$$

Only the original Table I monomials are differentiated, exactly once. No already differentiated graph, or graph involving X, w, z , is differentiated again. Thus no unspecified higher derivative enters the calculation.

Lemma 6.2 (Closure of the graph calculation). *The preceding rules evaluate all terms of $\mathcal{I}$ and Γ by a finite sum of scalar graphs. In every intermediate graph a free slot has exactly one incidence, each cubic tensor has exactly three incidences, and there are at most two cubic tensors. Every resulting scalar graph has degree two in (w, z, ℓ, X) .*

Proof. The incidence assertions hold for each table row by its decoding. A disjoint union preserves them. A permutation only renames free slots. Contracting two free slots removes their unique incidences and joins the remaining ends, or closes a trace; it does not change the number of slots of any cubic tensor. The derivative insertion described above replaces one edge by two incidences of a new X and supplies its third incidence as a new free slot. Trace differentiation does the same with a self-edge. The q_l term supplies two incidences on its self-edge and the third at l . Derivatives of p and g supply a free derivative slot without introducing a cubic tensor. These statements prove the incidence assertions by induction over the operations.

Table I has degree zero and its derivative has degree one in the indicated variables. The formula for $\mathcal{I}$ multiplies such derivatives by one further factor, or a degree-zero coefficient by two factors. Each Table II row has degree one; projection, contraction, and multiplication by χ preserve that degree. Squaring produces degree two. In particular no operation used here needs three copies of X . Termination is immediate from the finite tables, finite permutation sums, and the finite product rule; every contraction reduces the number of free slots by two. $\square$

The linear contribution of each Table I row is obtained by substitution into (4.3), forming $C = K - K^{(12)}$ for a K row. One check on the slot order and coefficients is the single row $U = A$ with coefficient one. Its interior contribution is

$$5z^2 + |X|^2 + (2a - 2)|q|^2,$$

and its boundary contribution is $2\lambda - \text{tr } A^3$. To expand a Gram block, sum over unordered pairs of rows s, t : the coefficient is $\sum_c c_{sc} c_{tc} / Q^2$, doubled when $s \neq t$, and the graph polynomial is the contraction of row t with Π applied to row s , multiplied by ρ . This uses self-adjointness of Π and accounts for every term of (6.2) exactly once.

6.4 Bounds for scalar graphs

Put $\theta = 4/5$. We first establish the tensor estimates that will bound each scalar graph, retaining the dependence between p and g .

Lemma 6.3. *For the constrained matrix A and symmetric trace-free tensor X ,*

$$\|A\|_{\text{op}}^2 \leq \theta, \quad |A^k|^2 \leq \theta^{k-1} \quad (k \geq 1), \quad (6.9)$$

$$|\text{tr } A^k|^2 \leq \theta^{k-2} \quad (k \geq 2), \quad |\text{tr } A^3|^2 \leq \frac{9}{20}, \quad (6.10)$$

$$|X : A|^2 \leq \frac{5}{7}|X|^2, \quad (X : A)_i = X_{ijk}A_{jk}. \quad (6.11)$$

If $h, b > 0$ and $g^2 + |p|^2 \leq 1$, then

$$|g|^{2h}|p|^{2b} \leq \left(\frac{h}{h+b}\right)^h \left(\frac{b}{h+b}\right)^b. \quad (6.12)$$

Proof. Let the eigenvalues of A be $\alpha_1, \dots, \alpha_5$. For each i ,

$$\alpha_i^2 = \left(\sum_{j \neq i} \alpha_j\right)^2 \leq 4 \sum_{j \neq i} \alpha_j^2 = 4(1 - \alpha_i^2).$$

This proves the operator bound, and $\sum_i \alpha_i^{2k} \leq \theta^{k-1} \sum_i \alpha_i^2$ proves the Frobenius bound. Moreover $|\sum_i \alpha_i^k| \leq \sum_i |\alpha_i|^k \leq \theta^{(k-2)/2} \sum_i \alpha_i^2$ for $k \geq 2$.

For the sharper cubic bound maximize and minimize $\sum_i \alpha_i^3$ on the compact set $\sum_i \alpha_i = 0$, $\sum_i \alpha_i^2 = 1$. The gradients of the two constraints are independent there. Lagrange multipliers give $3\alpha_i^2 = \mu + 2\nu\alpha_i$, so an extremum has two distinct eigenvalues (a single value would violate the constraints). If their multiplicities are $j, 5-j$, the values are, up to simultaneous sign,

$$\sqrt{\frac{5-j}{5j}}, \quad -\sqrt{\frac{j}{5(5-j)}}.$$

Their cubic trace has absolute value $|5-2j|/\sqrt{5j(5-j)}$. Its square is $9/20$ for $j = 1, 4$ and $1/30$ for $j = 2, 3$, proving (6.10).

To prove (6.11), fix a vector ζ and symmetrize $\zeta \otimes A$:

$$B_{ijk} = \frac{\zeta_i A_{jk} + \zeta_j A_{ik} + \zeta_k A_{ij}}{3}, \quad v = \text{Tr}_{12} B = \frac{2}{3} A \zeta.$$

Expanding the nine terms in $|B|^2$ gives $|B|^2 = |\zeta|^2/3 + 2|A\zeta|^2/3$. The symmetric trace-free projection is $F = B - \mathcal{J}v/7$, in the notation of Lemma 6.1. Since $|\mathcal{J}v|^2 = 21|v|^2$ and $\langle B, \mathcal{J}v \rangle = 3|v|^2$,

$$|F|^2 = |B|^2 - \frac{3}{7}|v|^2 = \frac{1}{3}|\zeta|^2 + \frac{10}{21}|A\zeta|^2 \leq \frac{5}{7}|\zeta|^2.$$

Symmetry and trace-freeness of X give $\langle X : A, \zeta \rangle = \langle X, F \rangle$; Cauchy–Schwarz and duality prove the claim.

Finally put $s = g^2, t = |p|^2$. For $h, b > 0$, the maximum of $s^h t^b$ on $s, t \geq 0, s + t \leq 1$ lies on $s + t = 1$. Differentiating $h \log s + b \log(1 - s)$ gives the unique interior maximum $s = h/(h + b)$, proving (6.12). If either exponent is zero, the corresponding bound is simply 1. $\square$

For a nonzero scalar graph key t of degree two, let

$$y = (|w|, |z|, |\ell|, |X|)^T$$

and let $i, j \in \{1, 2, 3, 4\}$ be its two degree-one types, counted with multiplicity and ordered so $i \leq j$. Let h be its g exponent and b its number of p endpoints. Define L_t^2 as a product of the following rational factors:

1. for each trace $\text{tr } A^k$, use θ^{k-2} , replacing this by $9/20$ when $k = 3$;
2. for an edge of power k between distinct vertices, use θ^k ;
3. for an edge between two slots of the same X , use $5/7$ when $k = 1$ and θ^{k-1} when $k \geq 2$;
4. if $h, b > 0$, include the right side of (6.12); otherwise include 1.

Here separate occurrences of a vector endpoint count as distinct vertices; thus a p – p edge is governed by Item 2, not Item 3. Power-zero self-edges at X have already vanished in the collection rules.

Lemma 6.4 (Complete graph bound). *Every scalar graph retained by Lemma 6.2 satisfies*

$$|\mathcal{M}(t)| \leq L_t y_i y_j \tag{6.13}$$

on $g^2 + |p|^2 \leq 1$.

Proof. First separate all scalar trace factors and use Lemma 6.3. A cubic tensor has at most one self-edge, since such an edge consumes two of its three incidences. An edge of power $k \geq 2$ reduces that tensor to the vector $v_i = X_{ijk}(A^k)_{jk}$, for which $|v| \leq |X||A^k| \leq |X|\theta^{(k-1)/2}$. For $k = 1$ use (6.11). These reductions leave at most two cubic vertices and no self-edge on either.

We list all possible connected components of the remaining graph. With zero cubic vertices, a component is one edge joining two vectors; its contraction is bounded by the operator norm of its matrix times the two vector norms. With one cubic vertex, all its three slots end at vector leaves. Move each edge matrix onto its vector, and use tensor Cauchy–Schwarz, obtaining the norm of X times the three vector norms and the product of the three operator norms.

If a component contains two cubic vertices, they are joined by $r = 1, 2$, or 3 edges; every other slot ends at a vector leaf. Contract these leaves first, obtaining rank- r tensors Y, Z with the same preceding norm bounds. Up to a permutation of their slots, the remaining contraction is

$$\left\langle Y, (A^{k_1} \otimes \cdots \otimes A^{k_r}) Z \right\rangle.$$

Its absolute value is at most $|Y||Z|\prod_{s=1}^r \|A^{k_s}\|_{\text{op}}$. This includes all three parallel edges when $r = 3$, without a dimension factor. If the two cubic vertices have no edge between them, they lie in two components of the one-vertex type already treated. These cases are exhaustive: all other vertices have valence one, so cannot occur internally on a path, and there are no further cubic vertices. The reduced vectors from self-edges are ordinary vector leaves in this classification.

Multiply the estimates over components. Each original ℓ or X norm appears once per occurrence, all ordinary edge matrices contribute $\theta^{k/2}$, and the p endpoints contribute $|p|^b$. The scalar factors g^h and $|p|^b$ are bounded jointly by (6.12). Finally the total degree is two, so all remaining w, z, ℓ, X factors give exactly $y_i y_j$. $\square$

We can now bound every term left after subtracting the nonnegative comparison forms. The graph estimates reduce this remainder to a quadratic form in four magnitudes. The next calculation shows that its error is smaller than the reserved positive diagonal terms.

6.5 The interior remainder and its coercive margin

Set

$$d = \left(\frac{2}{25}, \frac{2}{25}, \frac{1}{4}, 1 \right), \quad E = \Gamma + \sum_{i=1}^4 d_i y_i^2 - \mathcal{I}. \quad (6.14)$$

The squared norms in Equation (6.14) are polynomial contractions. Expand and collect $E = \sum_t e_t \mathcal{M}(t)$ by Section 6.3, discarding zero coefficients only after their signed contributions have been summed. All coefficients are rational. The denominator Q^2 suffices: Gram terms have that denominator, and the denominators of a, d_i , and all Table I coefficients divide it after multiplication by Q^2 .

For each retained key let n_t be the least nonnegative integer satisfying

$$n_t^2 \geq 10^{10} L_t^2, \quad \widehat{L}_t = \frac{n_t}{10^5}. \quad (6.15)$$

There is no floating-point square root in this prescription. For a rational $s = 10^{10} L_t^2$, compute $n = \lfloor \sqrt{s} \rfloor$ by integer arithmetic and increase it by one exactly when $n^2 < s$. Form a symmetric matrix D_0 by adding

$$\frac{|e_t| \widehat{L}_t}{1 + \mathbf{1}_{\{i \neq j\}}} \quad (6.16)$$

to its ij entry and also its ji entry if $i \neq j$, starting with zero. The factor $1/2$ in an off-diagonal entry is required because the associated term in $y^T D_0 y$ is $2(D_0)_{ij} y_i y_j$. Lemma 6.4 and the triangle inequality therefore give

$$|E| \leq y^T D_0 y. \quad (6.17)$$

For clarity, the exact finite calculation needed here is the following. Decode all rows of Tables I and II; expand the nineteen Gram blocks by unordered row pairs and the finite projections (6.4)–(6.6); subtract each Table I contribution to (4.3); and add the four contractions with coefficients d_i . Use the contraction, derivative, and key ordering rules above after each operation. Combine all signed coefficients at equal keys. For each nonzero key compute its rational L_t^2 , perform (6.15), and

add (6.16). This gives all ten upper-triangular entries as follows:

(i, j)	nonzero keys	$2 \cdot 10^5 Q^2 (D_0)_{ij}$
(1, 1)	89	321268243322644
(1, 2)	53	96616799870252
(1, 3)	123	294707669526536
(1, 4)	170	385792585960684
(2, 2)	64	269583307398596
(2, 3)	85	115717471031956
(2, 4)	106	125155603107368
(3, 3)	238	605509501361296
(3, 4)	414	618424270356936
(4, 4)	372	1004504025266456

(6.18)

In particular there are 1714 nonzero keys. These finite sums use only integer and rational operations and comparisons; the complete literal input is in Appendix A. The accompanying program `verification/checks/reproduce.py` implements the specified calculation and records the individual coefficients as well as these sums.

Entrywise, (6.18) implies $D_0 \leq \bar{D}$, where

$$\bar{D} = \frac{1}{10^5} \begin{pmatrix} 2571 & 773 & 2358 & 3087 \\ 773 & 2157 & 926 & 1002 \\ 2358 & 926 & 4845 & 4948 \\ 3087 & 1002 & 4948 & 8037 \end{pmatrix}. \quad (6.19)$$

For example, this entrywise check consists of comparing each last-column integer in (6.18) with the corresponding integer in (6.19) multiplied by $2Q^2$. Let $\nu = (2484, 1440, 996, 304)^T$. The exact diagonal slacks are

$$\left(d_i - \frac{(\bar{D}\nu)_i}{\nu_i} \right)_{i=1}^4 = \left(\frac{673}{18400}, \frac{438907}{12000000}, \frac{2844869}{24900000}, \frac{3479389}{7600000} \right). \quad (6.20)$$

For $y_i \geq 0$, weighted Young inequalities give

$$2y_i y_j \leq \frac{\nu_j}{\nu_i} y_i^2 + \frac{\nu_i}{\nu_j} y_j^2, \quad y^T D_0 y \leq y^T \bar{D} y \leq \sum_i \frac{(\bar{D}\nu)_i}{\nu_i} y_i^2.$$

Use $\Gamma \geq 0$ in (6.14) and then (6.17). It follows that $\mathcal{I}$ is at least the sum of y_i^2 times the four positive slacks in (6.20). In particular the coefficients of z^2 and $|\ell|^2$ are at least ϵ_* , proving (4.3). To obtain the stated positivity of f , keep A, p, g fixed, set $X = w = z = 0$, and choose $|\ell| = 1$. Then $q = 0$, every D term is zero, and $\mathcal{I} = f$.

6.6 Boundary reduction and the eleven coefficients

It remains to verify (4.4). Orthogonal equivariance allows us to take $p = e_5$ and to diagonalize A on $p^\perp$. Write

$$A = \text{diag}(x_1, x_2, x_3, x_4, \lambda), \quad \lambda = -\sum_{i=1}^4 x_i, \quad R(x) := \sum_{i=1}^4 x_i^2 + \left(\sum_{i=1}^4 x_i \right)^2 = 1. \quad (6.21)$$

No sign or distinctness condition is imposed on the x_i ; the normal eigenvalue condition is exactly $\sum_i x_i > 0$. In these coordinates $(S_0)_{55} = 0$ and $(S_0)_{\alpha\alpha} = b_\alpha := \lambda x_\alpha - x_\alpha^2$. The only nonzero entries of V are $V_{555} = 1$ and the three permutations of $V_{5\alpha\alpha} = b_\alpha$. Since $C_{55jk} = 0$, a direct component formula for the boundary polynomial is therefore

$$\mathcal{S} = -\lambda f + U_{55} + M_5 + \sum_{\alpha=1}^4 b_\alpha (U_{\alpha\alpha} + C_{5\alpha 5\alpha} + C_{5\alpha\alpha 5}), \quad (6.22)$$

where the fixed polynomials are evaluated at (6.21), $p = e_5, g = 0$. This formula does not assume additional slot symmetries of U or K .

Equivalently, one may contract the boundary expression as a scalar graph and then substitute

$$p \cdot A^k p = \left(-\sum_i x_i \right)^k, \quad \text{tr } A^k = \sum_i x_i^k + \left(-\sum_i x_i \right)^k. \quad (6.23)$$

For $k = 0$, the latter is 5 and the former is 1. The trace reductions $\text{tr } A = 0$, $\text{tr } A^2 = 1$ remain valid on $R = 1$. Thus either (6.22) or (6.23) evaluates the same scalar on all the allowed data. Orthogonal invariance under permutations of $p^\perp$ shows that $\mathcal{S}$ is symmetric in $x_1, \dots, x_4$.

Define the boundary remainder on this constraint set by

$$E_b = \Lambda + \frac{1}{4} - \mathcal{S}.$$

Expanding the two boundary Gram blocks and the Table I expression gives a polynomial representative with monomials of degree at most five. A finite way to check the degree assertion is to use (6.22): discard positive g powers, substitute the diagonal matrix and normal vector in each of the 231 rows, and add the displayed terms. All remaining monomials have degree at most five. Each boundary J_c has degrees one and two, so its squared terms have degree at most four, or five after the H weight.

The following convention gives a small exact list of its coefficients. For an ordinary monomial x^α of degree $j \leq 5$, replace it by

$$x^\alpha R^{\lfloor (5-j)/2 \rfloor}. \quad (6.24)$$

This preserves its value on $R = 1$ and gives degree four if j is even and degree five if j is odd. Perform the replacement before collecting exponents. Because R is symmetric, it commutes with permutation averaging. For a nondecreasing exponent vector β put

$$m_\beta(x) = \text{Av}(x^\beta).$$

Expand (6.24) and sum the coefficients of all exponent vectors whose nondecreasing rearrangement is β . Their sum is the coefficient of m_β after averaging: indeed $\text{Av}(x^\alpha) = m_\beta$ for every member α of that orbit, including when exponents repeat. Thus sorting exponents and summing coefficients introduces no extra factor of 24. Since $\mathcal{S}$ and the constant are symmetric and Λ was defined by averaging, the resulting averaged polynomial still equals E_b on $R = 1$.

Precisely, using the row-pair expansion of Section 6.3 for the two rank-1 blocks, subtracting the boundary contributions from Table I with denominator Q , and adding the homogenized constant $1/4$, one obtains

$$E_b(x) = \sum_{\beta} e_{\beta}^{(b)} m_{\beta}(x) \quad \text{on } R = 1, \quad (6.25)$$

with exactly the following eleven nonzero coefficients:

β	$Q^2 e_\beta^{(b)}$	β	$Q^2 e_\beta^{(b)}$
(0, 0, 0, 4)	−13479110	(0, 0, 0, 5)	17712712
(0, 0, 1, 3)	−93201124	(0, 0, 1, 4)	151038620
(0, 0, 2, 2)	−78543848	(0, 0, 2, 3)	244636976
(0, 1, 1, 2)	−139986162	(0, 1, 1, 3)	310091962
(0, 1, 2, 2)	445177656	(1, 1, 1, 1)	−4861942
(1, 1, 1, 2)	157995170		

(6.26)

This coefficient computation consists solely of the explicitly stated substitutions, multiplication of ordinary four-variable polynomials, and coefficient addition. It is also included in the accompanying exact reproducer.

On $R = 1$ every $|x_i| \leq 1$, hence $|m_\beta(x)| \leq 1$. Adding the eleven absolute integer coefficients gives

$$Q^2 \sum_{\beta} |e_\beta^{(b)}| = 1656725282.$$

Since $\sum_i x_i = -\lambda > 0$, (6.3) is nonnegative. It follows from (6.25) that

$$\mathcal{S} = \Lambda + \frac{1}{4} - E_b \geq \frac{1}{4} - \frac{1656725282}{62500000000} = \frac{6984137359}{31250000000} = \delta_*.$$

This completes the proof of Proposition 4.1. The estimate also persists at $\lambda = 0$ by continuity; repeated tangent eigenvalues cause no singularity in any of the formulas.

The pointwise comparisons and field bounds of Section 5 are therefore established. Section 7 now justifies their integration across Hessian zeros and near the suspension tips.

7 Hessian zeros, suspension tips, and the contradiction

We now justify the integration of the pointwise comparison. Retain the original smooth component $\Omega_o \subset S^{m_o}$, with $m_o \in \{4, 5\}$, from Proposition 2.4 and Lemma 2.5. Its tensor T_o is not identically zero. Write

$$B_o = \nabla T_o, \quad w_o = |T_o|, \quad \mathcal{Z}_o = \{y \in \overline{\Omega}_o : w_o(y) = 0\}, \quad \Omega_{o,+} = \Omega_o \setminus \mathcal{Z}_o.$$

The tensor T_o is smooth on the compact closure; its norm w_o is continuous there and smooth where it is positive. The working domain $\Omega \subset S^5$ is Ω_o when $m_o = 5$, and its suspension when $m_o = 4$. On the regular part of this working domain we use the unadorned notation of Sections 3 and 5. In particular, Z denotes the flux vector field, whereas $\mathcal{Z}_o$ denotes the original Hessian zero set. Set $M_o = \|w_o\|_{L^\infty(\Omega_o)}$. We fix

$$a = \frac{29}{50}, \quad \epsilon_* = \frac{438907}{12000000} > 0.$$

Interior and boundary integrals use their respective spherical volume and area measures; in the suspension, boundary integrals refer to its regular side boundary.

7.1 Weighted integrability on the original component

Lemma 7.1 (Weighted Bochner estimate). *The set $\mathcal{Z}_o$ is a compact subset of Ω_o , and*

$$\int_{\Omega_{o,+}} w_o^{2a-2} |B_o|^2 dV_o < \infty. \tag{7.1}$$

Consequently the function

$$D_o(y) = \begin{cases} w_o(y)^{2a}(1 + |B_o(y)|/w_o(y))^2, & y \in \Omega_{o,+}, \\ 0, & y \in \mathcal{Z}_o, \end{cases} \quad (7.2)$$

belongs to $L^1(\Omega_o)$.

Proof. The boundary conclusion of Lemma 3.2 gives $w_o > 0$ at every point of $\partial\Omega_o$. Compactness and continuity therefore give a boundary neighborhood on which w_o has a positive lower bound. The closed set $\mathcal{Z}_o$ is thus compactly contained in the interior. If it is empty, w_o has a positive minimum on the whole compact closure, and (7.1) follows immediately from the boundedness of B_o .

Otherwise choose $\eta \in C_c^\infty(\Omega_o)$ with $0 \leq \eta \leq 1$, equal to one on an open neighborhood of $\mathcal{Z}_o$. For $0 < \delta \leq 1$ put $W_\delta = w_o^2 + \delta$. Although w_o need not be smooth at its zeros, W_δ is smooth and strictly positive. The identities in Proposition 3.1 give

$$\Delta W_\delta = 2m_o w_o^2 + 2|B_o|^2 \geq 2|B_o|^2, \quad |\nabla W_\delta|^2 = 4|\langle T_o, B_o \rangle|^2 \leq 4W_\delta |B_o|^2.$$

Set $I_\delta = \int_{\Omega_o} \eta W_\delta^{a-1} |B_o|^2 dV_o$. Applying the chain rule to W_δ^a and integrating twice by parts against the compactly supported η yields

$$\begin{aligned} 2I_\delta &\leq \int_{\Omega_o} \eta W_\delta^{a-1} \Delta W_\delta dV_o \\ &= \frac{1}{a} \int_{\Omega_o} (\Delta \eta) W_\delta^a dV_o + (1-a) \int_{\Omega_o} \eta W_\delta^{a-2} |\nabla W_\delta|^2 dV_o \\ &\leq \frac{1}{a} \int_{\Omega_o} (\Delta \eta) W_\delta^a dV_o + 4(1-a) I_\delta. \end{aligned}$$

Since $2 - 4(1-a) = 4a - 2 = 8/25$, this proves the explicit uniform estimate

$$\frac{8}{25} I_\delta \leq \frac{1}{a} (M_o^2 + 1)^a \|\Delta \eta\|_{L^1(\Omega_o)}. \quad (7.3)$$

As $\delta \downarrow 0$, the nonnegative integrands restricted to $\Omega_{o,+}$ increase to $\eta w_o^{2a-2} |B_o|^2$ because $a < 1$. Monotone convergence and (7.3) control that integral. The compact set $\text{supp}(1-\eta) \subset \overline{\Omega}_o$ is disjoint from $\mathcal{Z}_o$; writing $b_\eta > 0$ for the minimum of w_o there, the remaining integral is bounded by

$$\int_{\Omega_{o,+}} (1-\eta) w_o^{2a-2} |B_o|^2 dV_o \leq b_\eta^{2a-2} \|B_o\|_{L^2(\Omega_o)}^2.$$

Thus all constants in this argument depend only on the fixed original component and its smooth data, on a , and on the chosen interior cutoff η , and are independent of δ . Finally, on $\Omega_{o,+}$,

$$D_o \leq 2w_o^{2a} + 2w_o^{2a-2} |B_o|^2.$$

The first term is integrable by compactness, and the second by (7.1). No measure, dimension, or regularity property of $\mathcal{Z}_o$ has been used. $\square$

7.2 Smooth angular cutoffs and the integrated comparison

Let $\vartheta \in C^\infty([0, \infty))$ satisfy $0 \leq \vartheta \leq 1$, $\vartheta = 0$ on $[0, 1]$, and $\vartheta = 1$ on $[2, \infty)$. Fix this profile once and for all, and set

$$L_\vartheta = \|\vartheta'\|_\infty, \quad \kappa = 2L_\vartheta.$$

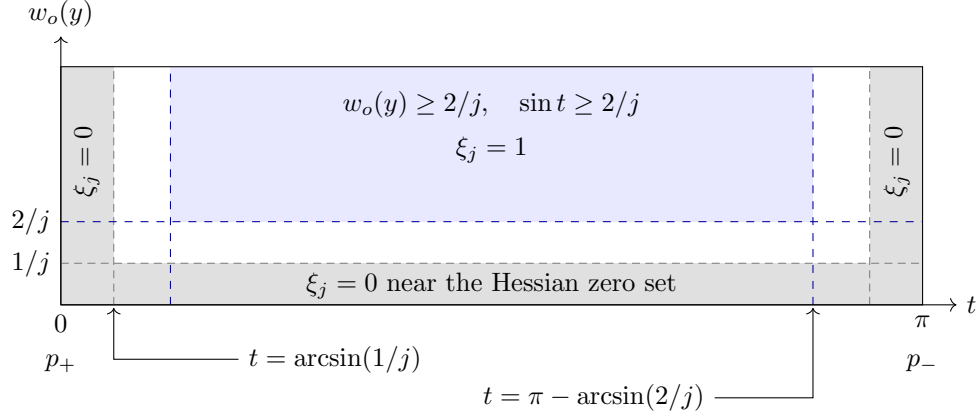


Figure 1: The product cutoff on a suspended component. The horizontal variable is the suspension angle in $\Phi(t, y) = (\sin t y, \cos t)$; the two end slices collapse to the tips p_+ and p_- . The vertical variable records the value of $w_o(y)$, not a coordinate on Ω_o . Gray regions have $\xi_j = 0$, and the blue region has $\xi_j = 1$; the intervening bands contain the smooth transitions. The scales are schematic, and no shape or regularity of the Hessian zero set is represented.

For integers $j \geq 3$ define

$$\alpha_j(y) = \vartheta(jw_o(y)), \quad \xi_j = \begin{cases} \alpha_j, & m_o = 5, \\ \alpha_j(y)\beta_j(t), & \beta_j(t) = \vartheta(j \sin t), \quad m_o = 4. \end{cases} \quad (7.4)$$

Here α_j is defined to be zero at $\mathcal{Z}_o$, consistently with the displayed formula. It is smooth on $\bar{\Omega}_o$: near any zero of w_o it is identically zero, and all its transitions occur where w_o is smooth and positive. In the suspension, β_j vanishes on whole neighborhoods of both tips.

Figure 1 displays the two exclusions made by the product cutoff in the suspended case. Its transition bands shrink toward the two tips and the original Hessian zero set. The weighted estimate above will control the errors produced in all these bands at once.

On $\Omega_+ = \{x \in \Omega : w(x) > 0\}$, use the fields

$$h = w^a \sqrt{f(A, p, g)}, \quad Z_l = w^{2a} (G_{lijk} X_{ijk} + w M_l + P_l)$$

constructed in Section 5. By Lemma 5.5, there is a fixed $C_F \geq 1$ such that

$$|h| \leq C_F w^a, \quad |\nabla h| \leq C_F w^a (1 + w + |X|), \quad |Z| \leq C_F w^{2a} (1 + w + |X|). \quad (7.5)$$

The constant depends only on a , the dimension 5, the finitely many coefficients defining f, G, M, P , and the positive lower bound $f \geq \epsilon_*$. In particular C_F is independent of the cone, w, X , and j .

For each fixed j , define $v_j = \xi_j h$ and $F_j = \xi_j^2 Z$ on Ω_+ , and extend them by zero where $w = 0$. These are smooth on the regular closure of Ω and are supported away from the tips when present. The same neighborhood argument as for α_j proves smoothness of these extensions; it does not require h or Z themselves to extend smoothly across $w = 0$. All products below containing one of these cutoff factors or its derivative are likewise understood to be zero on $w = 0$. The mass w^{2a} , whose exponent is positive, has its ordinary value zero there and involves no normalized tensor.

Lemma 7.2 (Integrated comparison). *For each fixed $j \geq 3$,*

$$\epsilon_* \int_{\Omega} \xi_j^2 w^{2a} dV \leq \int_{\Omega} (h^2 |\nabla \xi_j|^2 + 2 \xi_j h \langle \nabla h, \nabla \xi_j \rangle - 2 \xi_j \langle Z, \nabla \xi_j \rangle) dV. \quad (7.6)$$

Proof. The interior comparison of Proposition 5.2 gives

$$\int_{\Omega} \xi_j^2 (4h^2 + |\nabla h|^2 + \epsilon_* w^{2a}) dV \leq \int_{\Omega} \xi_j^2 \operatorname{div} Z dV.$$

Apply the divergence theorem to the smooth field F_j . In the suspension one may first truncate the domain inside the two neighborhoods where F_j vanishes; the artificial end boundaries have zero flux. On the regular side boundary the outward normal is $-N$. Thus the boundary comparison of Proposition 5.4, $N \cdot Z + Hh^2 \geq 0$, gives

$$\begin{aligned} \int_{\Omega} \xi_j^2 \operatorname{div} Z dV &= - \int_{\partial\Omega} \xi_j^2 \langle N, Z \rangle d\sigma - 2 \int_{\Omega} \xi_j \langle Z, \nabla \xi_j \rangle dV \\ &\leq \int_{\partial\Omega} H(\xi_j h)^2 d\sigma - 2 \int_{\Omega} \xi_j \langle Z, \nabla \xi_j \rangle dV. \end{aligned}$$

Corollary 3.6 applies to v_j and bounds the boundary integral by

$$\int_{\Omega} (|\nabla(\xi_j h)|^2 + 4\xi_j^2 h^2) dV.$$

Expanding the square and canceling the common terms gives (7.6). $\square$

Every cancellation above is made before removing the angular cutoff. For fixed j and radial index k , the tests $\zeta_k(r)v_j$ from Corollary 3.6 have compact support on the regular Euclidean cone. First take the variation parameter ε to zero on this fixed support; then let $k \rightarrow \infty$ in Equation (3.27) to obtain the constant 4. The smooth extension collar and the allowed variation size may depend on j and k . We now take $j \rightarrow \infty$ only in Equation (7.6), which contains the positive mass and three cutoff errors. The energy and boundary terms have already canceled for each fixed j .

7.3 Vanishing of all three errors

On $\Omega_{o,+}$, differentiation of α_j and $|\nabla w_o| \leq |B_o|$ give

$$|\nabla_o \alpha_j| \leq \kappa \frac{|B_o|}{w_o}. \quad (7.7)$$

To see the constant, wherever $\vartheta'(jw_o) \neq 0$ one has $1 < jw_o < 2$, so $j \leq 2/w_o$; outside this transition region the derivative is zero. The same argument gives

$$|\beta'_j(t)| \leq \frac{\kappa}{\sin t}. \quad (7.8)$$

At every fixed point of $\Omega_{o,+}$, $\alpha_j = 1$ and $\nabla_o \alpha_j = 0$ for all sufficiently large j . At every fixed $t \in (0, \pi)$ the corresponding statements hold for β_j .

Write $b_o = |B_o|/w_o$ on $\Omega_{o,+}$. Let $E_{1,j}$, $E_{2,j}$, and $E_{3,j}$ denote, in order, the three signed integrands on the right of (7.6).

First suppose $m_o = 5$, so $\Omega = \Omega_o$, and set $K_n = 1 + M_o$. Since $X = B_o/w_o$,

$$1 + w + |X| \leq K_n(1 + b_o), \quad |\nabla \xi_j| \leq \kappa(1 + b_o).$$

Equations (7.5) and the preceding bounds give

$$|E_{1,j}| + |E_{2,j}| + |E_{3,j}| \leq C_F^2(\kappa^2 + 4K_n\kappa)D_o. \quad (7.9)$$

Indeed the three coefficient bounds are respectively $C_F^2 \kappa^2$, $2C_F^2 K_n \kappa$, and $2C_F K_n \kappa$, and $C_F \geq 1$. The dominator is integrable by Lemma 7.1. Each error is eventually zero at every point of $\Omega_{o,+}$, and is defined to be zero on $\mathcal{Z}_o$. Dominated convergence therefore makes the right side of (7.6) tend to zero.

Now suppose $m_o = 4$. Write $s_t = \sin t$ and use the suspension coordinates $(t, y) \in (0, \pi) \times \Omega_o$. The exact formulas in Proposition 3.4 give

$$w(t, y) = \frac{w_o(y)}{s_t}, \quad |B(t, y)|^2 = \frac{|B_o(y)|^2 + 3 \cos^2(t) w_o(y)^2}{s_t^4}, \quad dV = s_t^4 dt dV_o. \quad (7.10)$$

In particular $\Omega_+ = (0, \pi) \times \Omega_{o,+}$, and there

$$|X| = \frac{\sqrt{b_o^2 + 3 \cos^2 t}}{s_t} \leq \frac{\sqrt{3}}{s_t} (1 + b_o).$$

The warped metric gives the exact cutoff identity

$$|\nabla \xi_j|^2 = \alpha_j^2 |\beta_j'|^2 + s_t^{-2} \beta_j^2 |\nabla_o \alpha_j|^2,$$

so (7.7)–(7.8) imply

$$|\nabla \xi_j| \leq \frac{\kappa}{s_t} (1 + b_o), \quad 1 + w + |X| \leq \frac{K_s}{s_t} (1 + b_o), \quad K_s = 1 + M_o + \sqrt{3}. \quad (7.11)$$

Multiplying the bounds for the three errors by the volume factor in (7.10) now gives

$$s_t^4 (|E_{1,j}| + |E_{2,j}| + |E_{3,j}|) \leq C_F^2 (\kappa^2 + 4K_s \kappa) s_t^{2-2a} D_o(y). \quad (7.12)$$

This inequality is initially read on $\Omega_{o,+}$, and both sides are assigned zero on $\mathcal{Z}_o$. Its constant depends only on the fixed certificate, a , ϑ , and M_o , and is independent of j , t , and y . It also shows explicitly why the original weighted $|B_o|^2$ estimate suffices: by (7.2),

$$s_t^{2-2a} D_o(y) \leq 2s_t^{21/25} (w_o^{29/25} + w_o^{-21/25} |B_o|^2) \quad \text{on } \Omega_{o,+}.$$

The right side is integrable with respect to $dt dV_o$, because $21/25 > -1$ and (7.1) holds. Every error is eventually zero at each fixed $(t, y) \in (0, \pi) \times \Omega_{o,+}$. Dominated convergence applied to (7.12) therefore makes the right side of (7.6) tend to zero also in the suspension.

Theorem 7.3. *The first dimension admitting a nonflat one-homogeneous global minimizer cannot be 5 or 6.*

Proof. If the first dimension were 5 or 6, the reductions would provide exactly one of the two working domains just considered. In either case, the right side of (7.6) tends to zero. On the original component, however,

$$0 < I_o := \int_{\Omega_o} w_o^{2a} dV_o < \infty.$$

Finiteness follows from boundedness and compactness. Positivity follows from $T_o \not\equiv 0$ and continuity, which give an open subset of Ω_o on which $w_o > 0$. In the nonsuspended case dominated convergence gives

$$\lim_{j \rightarrow \infty} \int_{\Omega} \xi_j^2 w^{2a} dV = I_o.$$

In the suspended case (7.10) gives instead

$$\lim_{j \rightarrow \infty} \int_{\Omega} \xi_j^2 w^{2a} dV = I_o \int_0^\pi (\sin t)^{4-2a} dt = I_o \int_0^\pi (\sin t)^{71/25} dt > 0.$$

Here the dominating mass density is the integrable, everywhere defined function $(\sin t)^{71/25} w_o^{29/25}$; its value is zero wherever $w_o = 0$. These strictly positive limits contradict (7.6), since $\epsilon_* > 0$. $\square$

The known lower-dimensional classification gives $d_* \geq 5$, and the dimension-seven globally minimizing example gives $d_* \leq 7$, as stated in Section 1. Theorem 7.3 excludes the only two remaining possibilities below 7. Hence $d_* = 7$, completing the proof of Theorem 1.1.

8 Regularity and sharpness of the dimension bound

We now pass from the classification of homogeneous global minimizers to local free boundaries. This step uses the dimension-reduction theorem of Weiss [18], in the formulation [17, Theorem 1.4 and Proposition 10.13].

Proof of Corollary 1.2. Apply the cited dimension-reduction theorem with volume coefficient one on balls compactly contained in D . Its inputs are uniform flatness regularity, linear nondegeneracy, compactness of blowups, and the fact that all blowup limits are homogeneous global minimizers. These are the constant-coefficient results used in Proposition 2.1; flatness regularity is [17, Theorem 8.1]. The theorem gives an empty singular set below d_* , local finiteness in dimension d_* , and Hausdorff dimension at most $n - d_*$ above it. Theorem 1.1 sets $d_* = 7$. Smoothness at each regular point follows from Proposition 2.1(iii). A countable cover by interior balls gives the assertions on D .

For sharpness, let U be the nonflat global minimizer in $\mathbb{R}^7$ of De Silva and Jerison [7, Theorem 1.1], and set $U_n(x, y) = U(x)$ on $\mathbb{R}^7 \times \mathbb{R}^{n-7}$. Repeated application of Proposition 2.6 shows that U_n is globally minimizing. At each point $(0, y_0)$, every rescaling of U_n is U_n itself, by homogeneity and translation invariance in y . It cannot be flat: a half-space normal would have zero y -component, forcing U to be flat. Thus $\{0\} \times \mathbb{R}^{n-7} \subset \text{Sing}(U_n)$. Its dimension is $n - 7$, and the proved upper bound gives equality. $\square$

A Complete coefficient tables

The following two tables specify the polynomials used in Section 6.1.1 and the comparison forms in Section 6.2. Every coefficient printed below is an integer numerator with denominator

$$Q = 250000.$$

Leading zeroes in monomial codes are significant. All rows and all block headers are included.

A.1 Decoding and block conventions

For nonnegative rank, read a monomial code in consecutive triples. Its first triple hbj contributes $g^h(\text{tr } A^3)^b w^{[j=1]} z^{[j=2]}$. Each subsequent triple uvk is an unoriented edge carrying A^k , with $A^0 = \text{Id}$. Endpoints $0, \dots, r-1$ denote the free slots in order, 5 denotes a copy of p , and 6 a copy of ℓ ; three occurrences of 7 are contracted with one X , and three occurrences of 8 with a second copy. Multiply all factors and contract all nonfree slots. There are no additional multiplicity factors; see Section 6.1.

In Table I the first field specifies the polynomial, the second its monomial, and the third its coefficient numerator. The free-slot orders are U_{jk} , K_{lijk} , M_l , and P_l , while f has rank zero. The numbers of rows for (f, U, M, P, K) are $(34, 57, 74, 32, 34)$, for a total of 231.

Each line of Table II beginning with @ starts a separate block and gives its rank, projection label, and multiplier mode. For each coefficient column, sum the monomials with the coefficients in that column divided by Q to obtain a tensor J_c . For nonnegative rank the block contributes $\rho \sum_c \langle J_c, \Pi J_c \rangle$ to Γ , where the modes I and B mean $\rho = 1$ and $\rho = \chi = 1 - |p|^2 - g^2$, respectively. The labels **none**, **anti**, **sym**, and **hook** specify exactly the operators of Section 6.2.1; they do not instruct any further normalization of the coefficients. Identical headers in different positions still define distinct blocks.

For rank -1 , the four code digits instead give the four individual exponents of x_1, x_2, x_3, x_4 . The block contributes to Λ the average over all 24 permutations of $\rho \sum_c J_c^2$. Here I means $\rho = 1$, while H means $\rho = \sum_i x_i = -\lambda$. The averaging and boundary normalization $R = \sum_i x_i^2 + (\sum_i x_i)^2 = 1$ are those of Section 6.2. Table II contains 21 blocks: 19 interior blocks with 529 rows and two boundary blocks with 14 rows each, hence 557 coefficient rows in total. Each block retains its own number of coefficient columns.

A.2 Table I: the multiplier coefficients

The columns below read from top to bottom, first on the left and then on the right.

f 000 172575762	f 200551552 -7335033
f 200 -265322011	U 000500510 209038273
f 400 301491831	U 200500510 -16767570
f 600 -44526835	U 400500510 -283063959
f 800 14565793	U 600500510 -30248846
f 000550 -54772531	U 000550500510 -136083135
f 200550 219932290	U 200550500510 -427782476
f 400550 52506144	U 400550500510 -31623665
f 600550 9630205	U 000550550500510 10893857
f 000550550 137906751	U 200550550500510 546331605
f 200550550 106469818	U 000550550550500510 427088076
f 000550550550 -131450161	U 000011 170760340
f 200550550550 46298458	U 200011 -1206356
f 000550550550550 19137894	U 400011 121148296
f 000551 119093719	U 600011 41094485
f 200551 -79672895	U 000550011 159740708
f 400551 -9715716	U 000500511 -12972332
f 000550551 -60378744	U 200550011 -371328968
f 200550551 14546676	U 200500511 158604412
f 400550551 -57731025	U 400550011 323446445
f 000550550551 156426055	U 400500511 -150167309
f 200550550551 -335208469	U 600550011 -71599174
f 000550550550551 -114479609	U 000550550011 -98427304
f 000552 36551855	U 000550500511 -66406904
f 200552 -61391750	U 000551500510 161528440
f 400552 -54261369	U 200550550011 360663252
f 000550552 47470176	U 200550500511 -17938869
f 000551551 38322040	U 200551500510 84273648
f 200550552 -123645111	U 400550550011 -202347674
f 200551551 21745017	U 400550500511 -24865978
f 000550550552 -27358875	U 400551500510 20187308
f 000550551551 -10633583	U 000550550550011 -95478557
f 000551552 -63061863	U 000550550500511 243609372

U 000550551500510 -260094552
 U 200550550550011 -175753712
 U 200550550500511 -115703729
 U 200550551500510 -18269297
 U 000550550550550011 52505668
 U 000550550550500511 -136286395
 U 000550550551500510 124924734
 U 000551011 -34002037
 U 200551011 -87426923
 U 400551011 -16110396
 U 000550551011 -51017991
 U 000551500511 182661478
 U 000552500510 -212756173
 U 200550551011 98857514
 U 200551500511 -82275068
 U 200552500510 -68411390
 U 000550550551011 -365742
 U 000550551500511 121156685
 U 000550552500510 65143542
 U 000551551500510 -118149741
 U 000551551011 20474336
 U 000552500511 13163650
 U 200551551011 19565853
 U 200552500511 -13163652
 U 000550552500511 -13163652
 M 000500 119808440
 M 200500 -78615281
 M 400500 58172228
 M 600500 117063991
 M 000550500 -8257673
 M 200550500 -246670620
 M 400550500 546437417
 M 000550550500 -304458502
 M 200550550500 147099465
 M 000550550550500 -259741510
 M 000501 13135177
 M 200501 230900882
 M 400501 503642015
 M 600501 113276750
 M 000550501 284895091
 M 000551500 -41618173
 M 200550501 895262372
 M 200551500 -64307015
 M 400550501 -501973998
 M 400551500 203881659
 M 000550550501 -251554675
 M 000550551500 -68782868
 M 200550550501 -641088126
 M 200550551500 179691016
 M 00055055050501 139784474
 M 000550550551500 -40969083
 M 000502 -24304787
 M 200502 -622936552
 M 400502 268923235
 M 600502 168388434
 M 000550502 -108635150
 M 000551501 318683040
 M 000552500 33990338
 M 200550502 361921940

M 200551501 153090065
 M 200552500 -747048669
 M 400550502 558178192
 M 400551501 -1665598002
 M 400552500 1296406233
 M 000550550502 411493367
 M 000550551501 -121808407
 M 000550552500 11773023
 M 000551551500 67252403
 M 200550550502 -446325581
 M 200550551501 79505560
 M 200550552500 125885807
 M 200551551500 -19251826
 M 000550550550502 -408356086
 M 000550550551501 2958960216
 M 000550550552500 -2765855048
 M 000550551551500 -39050858
 M 000503 31451284
 M 010500 -111968845
 M 200503 461057708
 M 210500 221824980
 M 400503 -412407021
 M 410500 -101679590
 M 000550503 121978129
 M 000551502 -465236572
 M 000552501 -85409256
 M 000553500 501586157
 M 010550500 189809541
 M 200550503 -457984830
 M 200551502 -131436746
 M 200552501 1156439419
 M 200553500 -438673045
 M 210550500 -199478216
 M 000550550503 -37100467
 M 000550551502 376993983
 M 000550552501 216559416
 M 000550553500 -236527028
 M 000551551501 118549720
 M 000551552500 -289837183
 M 010550550500 -92636597
 P 100500 311718636
 P 300500 -206555629
 P 500500 -10345809
 P 700500 69636002
 P 100550500 -335187565
 P 300550500 141480631
 P 100550550500 -48726916
 P 300550550500 -59898908
 P 100550550550500 -130179927
 P 700501 664922213
 P 100551500 37613817
 P 500550501 2068537473
 P 100550551500 -80492409
 P 300550550501 -3956993330
 P 300550551500 2941040012
 P 100550550550501 -4535791726
 P 100550550551500 2719912082
 P 100552500 59422497
 P 300552500 -112216670

P 100550552500 160359392	K 400500511231 209730769
P 100551551500 -210782200	K 000550500511231 -47161546
P 100551502 -955698012	K 000551500520131 126782841
P 100552501 289629865	K 000500512520530 -310431451
P 100553500 289477038	K 200550500511231 134860957
P 300551502 290419431	K 200551500520131 -72013133
P 300552501 -244526547	K 200500512520530 90786861
P 300553500 -89919120	K 000550550500511231 -155714998
P 100550551502 372091130	K 000550551500520131 4534988
P 100550552501 -196017256	K 000550500512520530 171186722
P 100550553500 -130830861	K 000551500511520530 -64357492
P 100551551501 223993677	K 000500512231 380733573
P 100551552500 -287868455	K 200500512231 -425302994
K 000500520131 -395492688	K 400500512231 164988206
K 200500520131 557130979	K 000550500512231 -325733710
K 400500520131 -208630461	K 000551500511231 -84676007
K 000550500520131 265035499	K 000552500520131 6581825
K 000500511520530 177271543	K 200550500512231 449829666
K 200550500520131 -288587200	K 200551500511231 7194909
K 200500511520530 -18030270	K 200552500520131 -6581826
K 000550550500520131 -29990138	K 000550550500512231 153898314
K 000550500511520530 -153812921	K 000550551500511231 46595618
K 000500511231 142329663	K 000550552500520131 -6581826
K 200500511231 -265174093	

A.3 Table II: the comparison coefficients

The block headers are part of the data.

@ 0 none I
001 12

@ 0 none I
002 184

@ 1 none I
000771700 -638806
000750750701 -983284
000750751700 1320922
000750750750501 805293
000750750701550 1754296
000600 503057
000600551 -1946630
000600550551 1923242
000651501 -2068563
000601551 -1021342
001502 -3593596
001550502 4994642

@ 1 none B
000602 1754266
000601551 -1522955
000602550 -2522248

@ 2 sym I
000750700710 1391786 1177679
200750700710 -635584 -1188129
000750750700510 -669474 -759226

200771700510 145856 439798
000771750500510 -365548 1663410
000771700550510 333085 -1276634
000750750750011 898927 557074
000750750700511 -311872 -891024
000750750701510 -41287 -1733462
000750751700510 -93961 1393977
000750700710551 113136 227458
000771750011 57792 -1227345
000771700511 -2146827 33344
000771701510 -214285 799004
200771750011 1112073 309389
200771700511 1556631 843960
200771701510 -294789 -594312
000771750550011 -13597 795429
000771750500511 -329731 1039673
000771751500510 -836011 1028480
000771700550511 216212 -603835
000771700551510 1434254 -827393
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000600510 -388425 -131346
000600550510 -477571 -239908
000650011 1059131 288654
000601510 252356 -522796
200650011 -1055033 202800
200601510 422375 425949
000650550011 -1586221 -893296
000650500511 393362 557037
000651500510 573436 -1067580
000600550511 -290759 380780
000600551510 -876849 -172942
000651011 16748 -630547
000601511 18986 878826
000602510 -32503 767092
200651011 -572999 -197116
200602510 -62643 182099
000651500511 -99714 99762
000652500510 1165974 -802821
000601551510 -1273792 -46716
000602550510 151518 380083
001500510 -818099 -26218
201500510 551865 2425759
201011 807336 716540
401011 -1809265 -594
001550011 1952714 -432372
001500511 -970041 4080275
201550011 -3494733 -1591876
201500511 -2942057 1178858
001550550011 -2197984 152320
001550500511 -1126597 -3610821
001551500510 1302795 2046516
201012 453912 -635196
001550012 802719 479394
001500512 1149385 1541894
001501511 -570358 863506
102550011 2753121 -35533
102551011 -7449352 0
102500512 712433 -1155568
102501511 -648523 1532248

@ 2 anti I

000750750700510 -971778 -593070 595799
000750750700511 1851218 390321 -1032828
000750751700510 -2221492 -384831 773149
000771700511 1671121 1696523 -468940
000771701510 1285532 -64095 -352045
200771700511 362947 -2582045 -2929678
200771701510 -181856 1690707 -1313034
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000771700550511 -1653489 -1171267 -3601252
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@ 2 sym I

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@ 2 anti I

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@ 2 sym B

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 000601510 -75331 228090 -522865 124191 -472481
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302011 -893937 -2987523 1695406 937674 871113
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102500511 -5673921 0 0 0 0

@ 2 anti B

000771700510 -470106 -1867823
200771700510 -1539213 -3753948
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```

A.4 Reproduction of the finite arithmetic

In the complete source directory accompanying this article, run `make -C build check`; dependencies and the preserved calculation records are in `verification/checks/README.md`. The entry point enables assertions, fixes $a = 29/50$, and checks source pins, literal table transcription, and fresh exact outputs from two interior expansions and the boundary, projection, and graph calculations. The interior implementations agree on every key, signed coefficient, magnitude assignment, rounded bound, and aggregate. The retained provenance records describe the second implementation as separately authored from the mathematical specification and frozen before comparison with the first engine's output. That specification already contained the claimed aggregate totals. These checks verify finite arithmetic; correspondence between formulas and code, and the analytic argument, require mathematical review.

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